Reduced coordinates for the Collatz map: exact per-step laws, anchor dynamics, and the limits of counting arguments for cycles

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Abstract

The Collatz map sends an odd number x to $3x+1$ with every factor of 2 divided out. Much of an orbit is bookkeeping: while $x+1$ is divisible by 4, each step trades one factor of 2 in $x+1$ for a factor of 3. We compress each such deterministic valuation run into a single block, producing a self-map F on states (ω, d) — an odd core prime to 3 and a depth; the states appear independently as the skeleton rows of Williams (2026) — whose convergence problem and cycle set are exactly those of the Collatz map. In these coordinates the local arithmetic admits exact laws, governed — apart from a 3-adic absorption term, which obeys its own exact law — by one 2-adic quantity, the *anchor* $M(\omega) = -2\log\omega/\log 9 \in \mathbb{Z}_2$: the exit valuation obeys the global law $s = 2 + v_2(d - M(\omega))$ on the lifting classes and is constant on the rest; the depth evolution closes exactly in the anchor displacement together with a stated 3-adic absorption law; and the anchor increment obeys an exact law modulo any power of 2, computable from graded residues of the state. A finite window of digits and stratum labels consequently decides each step in an error-free trichotomy, while a digit-budget accounting, stated as a heuristic, indicates that no bounded window can decide infinite horizons — localizing the difficulty in the digit supply of the anchors.

On the cycle side, a one-line elimination identity (the Böhm–Sontacchi cycle equation in block form) yields short rederivations of the classical exclusions at one, two and three blocks and of the classical trim uniform in the number of blocks p (Rozier 1991; Simons–de Weger 2005), whose constant degrades like $(\log_2 3)^{-p}$. We prove that this exponential loss is intrinsic to size arguments. An explicit family of configurations (*staircases*: divergent-orbit profiles bent into loops) meets every rotation's exact size condition with slack between two absolute constants at every period $p \geq 3$ ($p \geq 16$ by proof, $3 \leq p \leq 15$ by finite check; at $p \in \{2, 4\}$ size-passing configurations are exhibited, with no bracket claimed), and, with every cycle element at least a threshold X_0, at every period $p \geq p_1(X_0)$, with slack exceeding the trim's own exponent by at most an absolute constant. At that threshold scale the family realizes the minimax value of Rozier's 1991 extremal under every rotation's exact size condition, the sharpness Simons–de Weger expected in 2005; the analogue in the cycle-length parameter, and the conclusion that uniform cycle exclusion needs divisibility input, are Halbeisen–Hungerbühler's (1997).

Finally we state the equidistribution ('fair-coin') hypothesis implicit in the classical heuristics as a precise conjecture — that the block letters of uniformly sampled large starts,

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tallied within an exponent budget proportional to the start's bit length and for every admissible pair of budget and horizon rates, carry, at every finite block length, the frequencies of an exactly computable Bernoulli law — prove its consequences conditionally, exhibit an unconditional base case inside the digit budget, and report a calibration campaign whose four apparent anomalies all dissolved under controls. The main text tells each idea in words; the statements, proofs and checks are in the appendices.

Version note

v4, October 2026 (DOI [**v4 version DOI**]). It supersedes v3 (DOI 10.5281/zenodo.21730505, August 2026), v2 (10.5281/zenodo.21421120) and v1 (10.5281/zenodo.21273548, July 2026); the concept DOI 10.5281/zenodo.21273547 resolves to the latest version. v4 attributes the uniform trim (Theorem D.5) to Rozier and to Simons–de Weger, with finiteness at each period to Brox, and presents it as a rederivation; states the sharpness of that trim (Theorem D.6) at every period $p \geq 3$ and at every size threshold X_0 for $p \geq p_1(X_0)$, with its proof, in place of the assessment and status report of v3; cites the classical and contemporary sources of the reduced coordinates' objects and of the elimination identity; and corrects the attribution of Theorem C.3's unconditional bound at $\omega = 1$. v4 is also rewritten for a curious reader without training in mathematics: the main text carries the ideas in words, and the statements, with every proof, are in Appendices B–E. Its statements are therefore renumbered, and Appendix H maps v3's numbers to v4's. Outside work that cites this paper's statements by number, including the cycle ledger shared with Eric Merle, cites v3, which remains the pinned citation for that purpose. The history of every version, item by item, is at `paper/collatz-reduced-version-history.md` in the project record.

Author's note

Dear reader — I'm no mathematician, and I learned much of this paper's terminology during the process of formalizing it (p-adic valuations were completely new to me as a formal concept). It all began when I got hooked on a strange recursive pattern in the prime factors of a Collatz chain, and that idea pushed me into conversation with AI. We cleaned my idea up into the states (ω, d), with the primary goal of factoring as much deterministic behaviour as possible out of the question, in the hope of focusing attention where it matters. What these coordinates contribute is modest: an exact description of each step, a construction agreeing that Rozier and Simons–de Weger's limit on counting arguments against cycles cannot be substantially improved, and a precise statement of the 'fair-coin' assumption behind the classical heuristics. Because this work is AI-assisted and because I am not formally trained in the relevant mathematics, Appendix A describes the verification protocol used throughout the project. The complete research record is public at https://github.com/macindoe/collatz. — Ben

1 What this paper is about

Take a positive whole number. If it is even, halve it; if it is odd, multiply it by 3 and add 1. The Collatz conjecture says that from every starting number, repeating this rule eventually reaches 1. After an odd step the result $3x + 1$ is always even, so it is convenient to do all the halvings at once and look only at odd numbers:

$$T(x) = \frac{3x+1}{2^{v_2(3x+1)}} \qquad (x \text{ odd and positive}).$$

Here $v_2(n)$, the 2-*adic valuation* of n, is the number of times 2 divides n — equivalently, the number of trailing zeros of n written in binary. So T multiplies by 3, adds 1, and strips every trailing zero. This odd-to-odd map is Tao's Syracuse map Syr [4], and the Collatz conjecture asserts that every T-orbit reaches 1.

This paper has three purposes: to prove exact per-step laws for the dynamics that T induces on blocks of steps; to determine how far counting arguments can go toward excluding cycles (answer: classically, exponentially far in the number of blocks [10, 19]; and, as we prove, no farther: at every period $p \geq 3$, and at every size threshold X_0 for all periods $p \geq p_1(X_0)$); and to state precisely the statistical hypothesis that the classical heuristics leave implicit.

Three levels of description. The paper looks at the map at three levels, one at a time.

(1) *The map* T, one odd step at a time.

(2) *Blocks.* Long stretches of a T-orbit are forced: while $x+1$ is divisible by 4, the next step trades one factor of 2 in $x+1$ for one factor of 3, and a run of such steps ends in a single burst of halvings. Section 2 shows this on an example and compresses each run into one block.

(3) *States.* Each block is described by two numbers, a state (ω, d), and the blocks chain into a map F from state to state that loses nothing the conjecture depends on (Section 3).

The rest of the paper works at the third level: exact laws for one step of F (Section 4); what counting can and cannot say about cycles (Section 5); and the 'fair-coin' hypothesis (Section 6). Section 7 says what remains open, and Section 8 who did what.

What is new. Each item is explained in the section named, and stated formally in the appendix named.

(i) A reduced self-map F on the states (ω, d) — the skeleton rows of Williams [15], reached independently — with an in-paper proof of well-definedness, whose convergence problem and cycle set are exactly equivalent to those of T (Section 3; Appendix B).

(ii) A global anchor law for the exit valuation s, derived from one squaring lemma (Section 4; Theorem C.3).

(iii) Exact residue-class formulas for the depth evolution, including an explicit 3-adic absorption law (Lemma C.4, Theorem C.5).

(iv) An exact low-order law for the anchor increment and an error-free trichotomy deciding the next step from a finite residue window together with the step's stratum labels, with a digit-budget accounting (Theorems C.7–C.8, Heuristic C.9).

(v) A cycle elimination identity (the Böhm–Sontacchi cycle equation in block form), short rederivations of the classical small-cycle exclusions and of the classical uniform trim (Rozier; Simons–de Weger), and a proof of that trim's sharpness: an explicit family meeting every rotation's exact size condition at every period $p \geq 3$ and, for $p \geq p_1(X_0)$, with every element at least any size threshold X_0, where it realizes the minimax value of Rozier's extremal profile, showing that the exponential loss is intrinsic to size-counting (Section 5; Appendix D).

(vi) The equidistribution hypothesis stated exactly, its consequences proved conditionally, and a calibration record (Section 6; Appendix E).

What this paper does not claim.

- It does not settle the Collatz conjecture.
- That bounded windows of digits cannot decide unbounded horizons (Section 4) is the organizing heuristic, not a formalized theorem.
- The exclusion of cycles with one, two or three blocks, and the uniform trim, are not new: they are classical (Steiner; Rozier; Simons–de Weger; Hercher; finiteness at each period is Brox's). What is offered for them is new derivations.
- Every statement about the staircase concerns the reach of size arguments, and none bears on whether cycles exist. Nothing is stated at period $p = 1$; at $p \in \{2, 4\}$ only the existence of size-passing configurations is shown, with no bracket; no closed form for the threshold period $p_1(X_0)$ uniform in X_0 is claimed, and nothing below $p_1(X_0)$.
- The fair-coin hypothesis is a conjecture. No drift or contraction statement about orbits follows from window equidistribution, and the descent consequence is not stated (it is unconditional, and stronger, in Inselmann [5]). The hypothesis's exceptional set is a set of starting values of vanishing density, not a null set of orbits; it says nothing about any orbit's infinite tail, does not exclude individual staircase tails, and does not iterate.

How to read this paper. Each section opens with what it is about and why it matters, in words. Every informal statement in the main text names its formal statement in an appendix, where definitions, constants, proofs and record pointers are given in full: Appendix B (the reduced system), C (the exact laws), D (cycles) and E (the hypothesis). Appendix A states how the work was done and checked; Appendix F discusses related work in detail; Appendix G indexes every computational claim to the script that checks it; and Appendix H maps v3's statement numbers to this version's.

2 One rising run: trading 2s for 3s

What this section is about. Most steps of a Collatz orbit are forced. This section shows the mechanism on one example, then states it in general. Everything else in the paper is built on it.

Start from $x = 7$ and write each odd number of its orbit, and the number one above it, in binary:

step	x	x in binary	$x+1$ in binary	$x+1$ factored
0	7	111	1000	2^3
1	11	1011	1100	$2^2 \cdot 3$
2	17	10001	10010	$2 \cdot 3^2$
3	13	1101	1110	$2 \cdot 7$

The steps from 7 to 11 and from 11 to 17 are *rising* steps: $3x + 1$ is divisible by 2 exactly once, so $T(x) = (3x + 1)/2$, which is larger than x. The reason is visible in the last column. Adding

1 to both sides of $T(x) = (3x+1)/2$ gives

$$T(x) + 1 = \tfrac{3}{2}\,(x+1)\,:$$

each rising step removes one factor of 2 from $x+1$ and puts in one factor of 3. In binary, $x+1$ loses one trailing zero per step (`1000`, `1100`, `10010`); equivalently, x loses one trailing 1 (`111`, `1011`, `10001`). When only one factor of 2 is left ($x+1 = 18 = 2\cdot 3^2$), the trade cannot continue: $3\cdot 17+1 = 52$ is `110100` in binary, divisible by 4, and the step from 17 halves twice, down to 13.

In general (Proposition B.2): write $x+1 = 2^m u$ with u odd, so that x ends in exactly m ones in binary. Then the next $m-1$ steps of T are rising steps of this kind, each converting one factor of 2 in $x+1$ into a factor of 3, and the step after them ends in a single cascade of halvings, to the odd number

$$x_{\text{exit}} = \frac{3^m u - 1}{2^s}, \qquad s = v_2(3^m u - 1) \geq 1.$$

The whole run — m odd numbers, $m-1$ rising steps and one cascade — is a *block*; m is its *entry depth* and s its *exit valuation*. For $x = 7$: $m = 3$, $u = 1$, the block is $7, 11, 17$, and $3^3 - 1 = 26 = 2\cdot 13$ gives $s = 1$ and the exit 13. This is a classical observation (Steiner [8]; see also Rozier [9] and LaDue [14]), and it was also the author's starting point (Section 8).

The same block in the one-division map. Much of the literature uses the map $x \mapsto (3x+1)/2$ for odd x and $x \mapsto x/2$ for even x, which halves once per step (Terras [1]). In it, a block of entry depth m and exit valuation s is exactly m odd terms followed by s even terms: for $x = 7$, the odd terms $7, 11, 17$, then the even term 26, then the next block starts at 13. This dictionary (Appendix D, "The dictionary with m-cycles") lets results stated in that language be read in blocks.

3 Blocks become states

What this section is about. A block is determined by fewer numbers than it seems. This section compresses each block to a pair of numbers, its state, and states the theorem that the compression loses nothing the conjecture depends on. The formal versions are in Appendix B.

In $x+1 = 2^m u$, split off the factors of 3 of u: $u = 3^a\omega$ with ω not divisible by 3. The number ω is the *odd core*; it is divisible by neither 2 nor 3. The exit of the block depends only on $3^m u = 3^{m+a}\omega$, that is, on ω and on the *depth* $d = m + a$. The pair (ω, d) is the block's *state*. Different odd numbers share a state: those of the state (ω, d) are exactly $x = 2^{d-a}3^a\omega - 1$ for $0 \leq a \leq d-1$, and all of them exit at the same place (Proposition B.2(ii)–(iii)). For the state $(1, 3)$ they are 7, 11 and 17: the block of Section 2 consists of exactly the numbers of its own state.

The exit x_{exit} is again odd and begins the next block, whose state is found the same way. This defines the *reduced map* $F(\omega, d) = (\omega_+, d_+)$ from each state to the next; Definition B.1 gives it by explicit formulas. For $x = 7$ the states run

$$(1,3) \;\to\; (7,1) \;\to\; (1,2) \;\to\; (1,1),$$

the blocks beginning at 7, 13, 5 and 1.

Nothing is lost (Theorem B.3). $(1,1)$ is a fixed point of F, and the only number whose state is $(1,1)$ is 1. The Collatz conjecture holds if and only if every F-orbit from a valid state (one with ω odd, positive and prime to 3, and $d \geq 1$) reaches $(1,1)$, and the nontrivial cycles of T correspond one-to-one to the cycles of F other than $\{(1,1)\}$. Every question the conjecture asks can therefore be asked of F.

The same states, with their representatives and the cascade, appear independently as the skeleton rows of Williams [15]; Section 8 gives the correspondence.

4 The exact laws of one step

What this section is about. Once a block is a state (ω, d), three questions decide where the orbit goes next: how far the cascade falls (the exit valuation s); whether the next state picks up factors of 3 (the *absorption* a_+); and how deep the next block is. Each has an exact answer. All of them except the absorption are governed by one number attached to ω, its *anchor*; the absorption obeys its own exact law. The statements and proofs are in Appendix C.

The anchor. The 2-*adic integers* are binary strings that run on forever to the left, $\ldots b_3 b_2 b_1 b_0$, added and multiplied like ordinary binary numbers with the carries running leftward without end. Every odd number has an inverse among them, and one can take logarithms. For each odd ω the anchor $M(\omega)$ is the unique 2-adic integer with $9^{M(\omega)} = \omega^{-2}$; in the 2-adic logarithm, $M(\omega) = -2\log\omega/\log 9$ (Definition C.1). For example $M(1) = 0$, and $M(\omega)$ is even exactly when $\omega \equiv \pm 1 \pmod 8$.

The valuation law (Theorem C.3). Call a state *lifting* when $3^d\omega \equiv 1 \pmod 8$, that is, when $\omega \equiv 1 \pmod 8$ with d even, or $\omega \equiv 3 \pmod 8$ with d odd. On the lifting classes,

$$s = 2 + v_2(d - M(\omega)) \;\geq\; 3,$$

and $v_2(d - M(\omega))$ is the number of lowest binary digits on which the depth d and the anchor $M(\omega)$ agree, counted from the right up to the first disagreement. Think of d as the reading of a binary counter and $M(\omega)$ as a fixed reference pattern: the length of the cascade is set by how many low-order bits of the counter match the pattern. On the other six classes s is a constant, 1 or 2:

	$\omega \equiv 1$	$\omega \equiv 3$	$\omega \equiv 5$	$\omega \equiv 7 \pmod 8$
d even	lifting: $s \geq 3$	$s = 1$	$s = 2$	$s = 1$
d odd	$s = 1$	lifting: $s \geq 3$	$s = 1$	$s = 2$

For $\omega = 1$ the anchor is 0, so on its lifting class (d even) $s = 2 + v_2(d)$: the state $(1,2)$ of the example has $s = 3$, and indeed $3 \cdot 5 + 1 = 16$ halves $1 + s = 4$ times, down to 1.

The factor of 3 (Lemma C.4). The next state gains a factor of 3 exactly when s is even. How many factors it gains is given by an exact 3-adic formula, which in one boundary case depends on ω through its expansion in powers of 3.

The next depth (Theorem C.5). On every residue class the next depth is $d_+ = m_+ + a_+$, with the absorption a_+ given by Lemma C.4 and the next entry depth m_+ given by an exact formula in the anchor displacement (the difference between the depth and the anchor) and s; the formula forces $m_+ \geq 1$ structurally.

How many digits one step needs (Theorems C.7 and C.8). The anchor is an infinite string, but one step needs only finitely many of its digits. Write $\sigma = s + m_+$. For every $k \geq 1$, the change of anchor $\Delta M = M(\omega_+) - M(\omega)$, read modulo 2^k (its lowest k binary digits), is determined by, and explicitly computable from, $\omega \bmod 2^{\sigma+k+2}$, $d \bmod 2^{\sigma+k}$ and $a_+ \bmod 2^k$; the stratum data (s, σ) are determined by the same residues, but the absorption is not — it is a 3-adic function of ω, and it enters only through its residue $a_+ \bmod 2^k$. Call these residues, together with the stratum labels (s, σ, a_+), the *depth-k window*. From the window alone one can determine the exact next depth and the next residue-parity class; the exact next exit valuation $s_+ \in \{1, 2\}$ if the next class is not lifting; and, if it is lifting, either s_+ exactly or the correct report "$s_+ \geq k + 2$". The window never outputs a wrong value: it either answers or says, correctly, that the next cascade is deep. It is undecided exactly when $s_+ \geq k + 2$; under the model law of Section 6 (with cap $D \geq k + 2$) that event has probability $2^{-(k+1)}$ exactly, and along real orbits the record measures an undecided rate of 0.0275 against 0.03125 at $k = 4$ and 0.0019 against 0.00195 at $k = 8$.

The digit budget (Heuristic C.9; a heuristic, not a theorem). Deciding one step at depth k uses the state's binary digits to depth $\sigma + k + 2$, where σ has mean 4 under the model law of Section 6. Theorem C.7 shows that these digits suffice; that they are also needed is part of the heuristic. Nothing regenerates them: the anchors of later states are functions of the full initial state, not of any bounded window. A window of W digits therefore supports $O(W/(k+6))$ decided steps. This strongly suggests — and the paper treats it as the organizing heuristic, not a formalized theorem — that bounded-window determinism cannot decide unbounded horizons without additional statistical or rigidity input.

Checks. The valuation, entry-depth and absorption laws were checked by independently written code, with zero discrepancies (Remark C.6), and the trichotomy along real orbits, with zero errors (Remark C.10); Appendix G lists the scripts.

Where the difficulty is. Every quantity above other than the absorption a_+ obeys an exact law in the anchor displacement $d - M(\omega)$, and a_+ obeys its own exact 3-adic law (Lemma C.4); a finite window of digits and stratum labels decides the next step in a trichotomy that never errs; and the digit-budget accounting, stated as a heuristic, indicates that each decision *consumes* digits irreversibly. In these coordinates the difficulty of the problem is not diffuse: it is the question of where the digits of the anchors come from. Heuristic C.9 is the paper's organizing negative observation: it splits the residual difficulty into a statistical question for typical starting values (Section 6) and a rigidity question for cycles (Section 5); we know of no residual content outside those two.

5 Cycles: what counting can and cannot do

What this section is about. A counterexample to the conjecture would be either an orbit that grows forever or a cycle that never reaches 1. Cycles are the one regime where finite data can beat unbounded depth: a cycle is a finite object, and arithmetic can be brought to bear on it. This section explains what the classical counting ('size') arguments achieve against cycles, and proves that they go no further: at every period $p \geq 3$, and at every size threshold X_0 for all

periods $p \geq p_1(X_0)$, the size conditions are met at the scale the classical bound (the 'trim') allows. Appendix D has the statements and proofs.

A cycle as numbers. Suppose F has a cycle of p blocks, with entry depths m_t and exit valuations s_t ($t = 0, \ldots, p-1$, read cyclically). Then $n = \sum_t m_t$ counts the multiplications by 3 around the cycle, $K = n + \sum_t s_t$ counts the halvings, and we write $q = 2^K - 3^n$. In the one-division map such a cycle has exactly p maximal runs of odd terms, and its local minima are exactly its p block entries; so a p-block cycle is what Simons and de Weger call an m-cycle with $m = p$ [19], and p is Rozier's number of local minima [10]. Every exclusion proved in that language therefore transfers: Hercher's theorem [20] excludes every nontrivial cycle of F with $p \leq 91$ blocks. Such exclusions bound the number of blocks only, and say nothing about how many blocks have depth $m_t \geq 2$.

Size. Going round the cycle multiplies by 3 exactly n times and divides by 2 exactly K times, adding small amounts along the way; returning to the start forces 2^K to exceed 3^n by only a little. Precisely (Corollary D.2), $2^K = 3^n \prod_t (1+\chi_t)$ with each $\chi_t > 0$ smaller than the reciprocal of a cycle element, so that if every cycle element exceeds X then $0 < K \log 2 - n \log 3 < p/X$. Size arguments are arguments from how close 2^K must be to 3^n. A convenient measure of that closeness is $\gamma := K - \log_2 q$: since $3^n/2^K = 1 - 2^{-\gamma}$, a large γ means that 3^n falls short of 2^K by the small fraction $2^{-\gamma}$.

Size and divisibility (Proposition D.1). The elimination identity — the cycle equation of Böhm and Sontacchi [11] in block form — turns a cycle into p conditions, one for each *rotation* r (each choice of starting block): an explicit positive integer R_r, built from the m_t and s_t, equals q times a positive integer. Hence $q \leq R_r$ for every r (the *size conditions*) and q divides every R_r (the *divisibility conditions*). Any tuple of positive integers m_t, s_t with $q > 0$ is a *configuration*, and one meeting every size condition is a *size-passer*. A cycle gives a size-passer; a size-passer need not give a cycle.

Small periods (Theorem D.4). F has no nontrivial cycles of 1, 2 or 3 blocks. These statements are contained in the classical results [8, 9, 19, 20]; the derivations, not the theorems, are the contribution.

The uniform trim: a short history. In 1991 Rozier bounded, for every number of local minima of a cycle — our p — the number of odd steps, with the exponent $(\log_2 3 - 1)/((\log_2 3)^p - 1)$, and observed that his bound is optimal over real run-length tuples, the extremal tuple growing geometrically at ratio $\log_2 3$ [10]. In 1997 Halbeisen and Hungerbühler, working in the cycle-length parameter, proved the size criterion optimal, attained by explicit rational cycles of every length, and reduced the remaining question to divisibility [12]. In 2000 Brox proved finiteness for each p [13]. In 2005 Simons and de Weger made the bound explicit, computed with it, and expected the exponential dependence on the number of minima to be unavoidable [19]. In this paper's notation the bound — the *uniform trim* — reads (Theorem D.5): every nontrivial p-block cycle satisfies

$$\gamma + \log_2 p \;>\; \kappa_p \, n, \qquad \kappa_p := \frac{\log_2 3 - 1}{(\log_2 3)^p - 1}.$$

The right side grows with the number n of rises, so a cycle with many rises needs 2^K very close to 3^n; but κ_p shrinks like $(\log_2 3)^{-p}$, so the requirement weakens exponentially as the number of blocks grows. Combined with Rhin's effective bound on how close 2^K can come to 3^n [24], it leaves at every period a finite, explicitly bounded set of candidates, $n \le n_0(p) = O(p\,(\log_2 3)^p)$. The argument uses only the size conditions, so the first inequality holds for every size-passer. Our proof is a rederivation by another route, chosen because it identifies the extremal profile that the staircase realizes.

The staircase, as a picture (Theorem D.6(iv)). At $p = 7$ and $n = 94$ take entry depths and exit valuations

$$m = (4, 7, 9, 15, 23, 35, 1), \qquad s = (1, 1, 1, 1, 1, 1, 49),$$

so that $K = 149$. The first six blocks climb, their depths growing roughly geometrically, each with the smallest possible exit valuation, 1; the last block has depth 1 and exit valuation 49, one long fall. This configuration passes all seven size tests with $\gamma = 6.74$, where the period-3 constant would demand $\gamma > 8.9$, and 84 further instances occur at $p = 6$ alone; all of them fail the divisibility conditions. The staircase profile — geometric depth growth with shallow exits — is exactly the growth regime of divergent orbits, bent into a loop by one crash. Size analysis cannot forbid it as a cycle for the same reason drift analysis cannot forbid divergence (Remark D.7): the two halves of the residual difficulty meet in one configuration, so the cycle half and the divergence half are, in a concrete sense, one object.

Sharpness (Theorem D.6). The staircase works at every period $p \ge 3$, and at every size threshold X_0 once $p \ge p_1(X_0)$.

- *At every period* $p \ge 3$ (part (i)). For every $p \ge 16$ there is a p-block size-passer with n between $(\log_2 3)^p$ and $1.05\,(\log_2 3)^p$, $K = \lceil n \log_2 3 \rceil$, and $3.683012 \le \gamma \le 5.140212$: slack between two absolute constants. For $3 \le p \le 15$ size-passers in the same bracket are found by finite check; at $p \in \{2, 4\}$ size-passers are exhibited, with no bracket claimed; nothing is stated at $p = 1$. Consequently, any constant c_p with $\gamma + \log_2 p > c_p\, n$ for every size-passer at a period $p \ge 16$ has $c_p < (5.140212 + \log_2 p)\,(\log_2 3)^{-p}$: the exponential loss in the trim is intrinsic to size arguments.

- *With every element large* (parts (ii)–(iii)). Every odd number below the verified range 2^{71} reaches 1 [25], so the elements of a nontrivial cycle all lie above it, and the natural test of size arguments puts every element above a threshold X_0. Every configuration has $\gamma - 1 \le \log_2(x_{\min} + 1) < \gamma + \log_2 p$, with $x_{\min}$ its least element; so a size-passer whose elements are all at least X_0 has $\gamma > \log_2(X_0+1) - \log_2 p$, and $\gamma \ge \log_2(X_0+1)+1$ is enough to put every element at least X_0. For every real $X_0 \ge 1$ and every period $p \ge p_1(X_0)$ there is a p-block size-passer with every element at least X_0 and $\lambda_0 \le \gamma - \kappa_p n \le \lambda_0 + 1.5$, where $\lambda_0 = 2.7095122\ldots$: slack exceeding the trim's own exponent by at most an absolute constant. The threshold period $p_1(X_0)$ is explicit through the continued fraction of $\log_2 3$: 10 for $X_0 \le 16$, 22 for $2^8 \le X_0 \le 2^{12}$, 36 at 2^{20}, 98 at Simons and de Weger's $X_0 = 5 \cdot 2^{60}$, 110 at 2^{71} and 309 at 2^{200}. No closed form uniform in X_0 is claimed, and nothing is claimed below $p_1(X_0)$.

- *What is attained* ("The threshold, explicitly", Appendix D). At that threshold scale the family realizes the minimax value of Rozier's extremal under every rotation's exact size

condition — that value, not the inequality of his Lemme 4 — the sharpness Simons and de Weger expected. There it attains the exponential dependence on the number of minima, and shows that the prefactor $p\,c_p$ of Simons and de Weger's Lemma 7 cannot be lowered below 0.054. Whether it can be lowered *to* a constant is left open, as Simons and de Weger leave it.

What is left. Uniform cycle exclusion therefore requires the divisibility conditions $q \mid R_r$ of Proposition D.1, not size alone — the conclusion Halbeisen and Hungerbühler reached in the cycle-length parameter [12]. Every statement in this section concerns the reach of size arguments, and none bears on whether cycles exist. The configurations of part (i) computed in the record all fail the divisibility conditions, as those of part (iv) do; those of part (iii) were not tested. Each part was checked by independently written code with exact integer arithmetic at every size decision (Appendix D, "Verification and record"; Appendix G).

6 The fair-coin hypothesis

What this section is about. The classical heuristic picture of Collatz orbits (Terras [1]) treats the halvings as tosses of a fair coin. This section states that picture exactly: the folklore heuristics become an exact statement about the block letters of typical orbits, equivalently about an explicitly computable window law. It then says where the statement is already a theorem, what it would and would not give, and how it was tested. Appendix E has it in full.

Letters and the model law. Read each block as a *letter* (m, r): its entry depth m and its exit valuation r. The model treats m and r as independent counts of the kind 'toss a fair coin until it lands tails', each value $j \geq 1$ having probability 2^{-j}; so a letter (m, r) has probability $2^{-(m+r)}$, and a word of letters the product of its letters' probabilities. On the reduced states this model becomes an exactly computable law $\pi_{k,D}$ on capped windows of the state (Appendix E); its 3-adic part, the stationary law of the absorption a_+, is not new here — it is Tao's Syracuse random variable [4], in the present normalisation. Under $\pi_{k,D}$, unconditionally, $P(s = j) = 2^{-j}$ at every j below the cap D; the rate at which a factor of 3 is gained is $\frac{1}{3}$, up to the cap; and the classical negative drift holds — but only as properties of $\pi_{k,D}$. Window equidistribution at each fixed (k, D) does not control the means of the unbounded m_+ and s, so no drift or contraction statement about orbits follows from it. The folklore 'ledger' of these facts is thus a theorem *about* $\pi_{k,D}$; the empirical question is only whether the orbits of typical starting values follow $\pi_{k,D}$.

The hypothesis (Hypothesis E.1). Pick a large N and draw an odd starting value x uniformly from $[N, 2N)$; write $b = \lfloor \log_2 N \rfloor$. Follow the blocks of x for a horizon of $H_N = \lceil \theta b \rceil$ blocks, and tally each block's letter only while the number of factors of 2 divided out so far is below a budget $E_N = \lceil \tau b \rceil$; both are proportional to the start's bit length, and later blocks are tallied as a blank †. The hypothesis says that for every finite word of letters and every $\varepsilon > 0$, the proportion of starts whose frequency of that word differs from its model value by more than ε tends to 0 as $N \to \infty$ — for every admissible pair (τ, θ) of budget and horizon rates. Admissible means two things, stated precisely in Appendix E: the budget is *protected* (all but a vanishing density of starts keep every in-budget block exit above any level X_N with $\log X_N = o(\log N)$),

and the pair is *consistent* (the model's own mean spending, 4 factors of 2 per block, fits the horizon inside the budget: $4\theta < \tau$).

Where it is already a theorem. The first blocks of a start are fixed by its own binary digits: the starts whose first blocks follow a prescribed itinerary form exactly one residue class (the classical coding fact, Terras [1]). So the first blocks of a start drawn from a window of 2^J consecutive integers are distributed exactly as the model says on the event that they have not divided out more factors of 2 than the start has binary digits (precisely, $S + 1 \leq J$, with S the factors of 2 divided out), and the error from that event and from the sampling window is explicitly bounded (Appendix E). Assembled with concentration estimates, this makes Hypothesis E.1 a *theorem* for every admissible (τ, θ) with $\tau < 1$, equivalently for every horizon rate $\theta < 1/4$, at every block length; the assembly is in the project record (`aeh.md` Lemma 13.2.4). The natural clock here is total exponent, not blocks: the frontier is one factor of 2 per bit of start — the start's own digit supply — and it reads as $1/4$ block per bit only through the model's mean spending of 4. The hypothesis is the assertion that the law still describes the orbit after the start's digits have been spent; the digit budget of Section 4 locates the frontier, and the hypothesis is the statistical statement on the far side of it. That frontier is the classical one — $\theta < 1/4$ is where Terras's count stops, and Korec's exponent [3] is obtained by exhausting it — and it is a frontier of this technique and not of the problem: Inselmann [5] crosses it unconditionally for the trajectory envelope and the first moment, in step time and by a stronger density notion rather than a sharper count.

What it would give, and what it would not. Under the hypothesis, for every ε and every horizon rate, all but a vanishing density of the odd starting values of each size carry the model's frequencies along their first H_N blocks, all but $o(H_N)$ of them within the budget, and all of them when $\tau < 1$; evaluating each odd x once, at its own dyadic scale, the starts that fail have natural density zero (record, `aeh.md` Proposition 13.2.5). What the hypothesis supplies beyond the classical count is distributional, and is the whole of its content: the full 2^{-j} law of s at every j below the cap, the exact $\frac{1}{3}$ rate of factor-3 gains, and the depth law, at every finite depth and past the budget at which the classical count stops. It does not control the union of its exceptional sets over all sampling scales. Its exceptional set is a set of starting values of vanishing density at a prescribed finite horizon — not a null set of orbits, and not a statement about any orbit's infinite tail. It does not exclude individual staircase tails; it does not iterate (the image of a density-one set of starts need not be density-one at the next scale, an obstruction Tao states in the same terms [4]); and, by Heuristic C.9, it is not expected to be reachable by finite-window computation. The descent consequence is not stated, because it is a theorem without the hypothesis, and a stronger one (Inselmann [5]). The hypothesis's content is a question about digits of 2-adic logarithms, older and broader than the Collatz problem.

How it was tested. An eight-round calibration campaign produced four apparent violations of the model, each dissolved by a sharper control: an excess at $s = 3$ and pair correlations, both artifacts of small starting values; digit biases from contamination by the 'bottom regime' of small integers, reaching $z = 41$ but reflecting particular numbers rather than a measure; and a 5σ digit bias in one cell, surviving a 2^{40} cut, which dissolved under fixed-horizon unweighted sampling (0.2503 ± 0.0015 against the null 0.25 over 84,739 visits) and was explained exactly by a deterministic routing of residues (Lemma E.2) that ratio estimators inherit. No residual

discrepancy was detected at any tested depth or cell, under the stated protocol and within three limits the campaign does not reach past: it tests block lengths 1 and 2 only; its adjudicating runs measure the across-orbit average of the per-start frequencies, not the per-start statement itself; and its altitude guard binds at finite size (Appendix E, "Calibration").

7 What remains open

In reduced coordinates the Collatz problem factors cleanly. The local arithmetic is under exact control: one 2-adic logarithm governs the step, apart from the 3-adic absorption, which has its own exact law, and finite windows decide steps without error. What remains is a single question wearing two costumes: for typical starting values, whether the anchors' digits equidistribute (Hypothesis E.1 — statistics); for cycles, whether the divisibility conditions $q \mid R_r$ can be met, which size alone cannot decide (Theorem D.6 — arithmetic). The staircase shows the costumes cover the same body. Three directions are open: arithmetic statements about cycles beyond the size level, which Theorem D.6 shows any uniform exclusion needs — for instance rigidity of closed anchor walks, since a cycle's anchors return and its increments satisfy $\sum_t \Delta M_t = 0$; transfer-operator analysis of the chain of exact depth-k windows (Theorem C.8); and any progress on digits of 2-adic logarithms of integers, a problem this paper inherits rather than creates.

8 Credit and provenance

The classical source of each result is named where the result is stated. The density results, the 3-adic comparisons and the cycle literature are discussed in Appendix F.

Two contemporary human–AI efforts are close in spirit: a documented exploration [28] develops a burst–gap decomposition and an orbit equidistribution conjecture, and a Lean 4 formalization [26] proves cycle exclusion conditionally on Baker-type and verification hypotheses, including an impossibility lemma for measure-based uniform bounds kindred to our Theorem D.6; neither contains the anchor machinery or the exact per-step laws. A bachelor's thesis of Rackl [27] (Klagenfurt, 2021) applies valuation methods to Syracuse cycles via the classical cycle equation; it likewise contains none of the present apparatus.

The division of labor in the present work is stated precisely, because it matters for both credit and blame. The mechanism underneath the reduced coordinates — each rising step trades a factor of 2 for a factor of 3 — comes from the author's hand-drawn note of March 2023, and is classical (Steiner [8]; see also Rozier [9] and LaDue [14]). The reduced coordinate itself, the factorization $x + 1 = 2^m 3^a \omega$ with its odd core ω, was worked out by the author by hand in December 2025, and was named and formalized as the state (ω, d) in AI-assisted work: with ChatGPT (OpenAI) from December 2025, and with Claude (Anthropic) from March 2026. The same states, with their representative fibre and intra-block cascade, appear independently as the skeleton rows (λ, k) of Williams [15], posted earlier, with $\lambda = \omega$ and $k = d$: Williams's Proposition 3.3, Theorem 3.6 and Proposition 3.10 are our Proposition B.2(iii), (i) and (ii). Her open problem (1) [15, p. 18] asks for the map sending one skeleton to the next, and Section 4 answers it in part: it gives exact laws for where a trajectory leaving a row lands — the exit valuation and the entry position, hence the row entered — but none for which skeleton is entered, where F restates Williams's Proposition 3.10. The analytical superstructure erected on the coordinates — the anchor formulation, the cycle apparatus, the equidistribution formalization,

and the proofs — was developed in AI-assisted research under the author's direction, drawing on the classical literature's concepts. Appendix A states the responsibility and verification protocol.

A Methodology, responsibility, and records

This work runs from a hand-drawn origin note (13 March 2023) through seventy-nine drafts to a research wiki whose pages carry status headers, verification records, and refuted claims with their refutation data. It was developed through extensive AI-assisted research in two phases. From December 2025 to March 2026 the author worked with ChatGPT (OpenAI): exploratory definitions, the early CollatzScout drafts, and, alongside them, the author's own hand computations in spreadsheets. From March 2026 the author worked with Claude (Anthropic). The author is responsible for the published claims, the research direction, and the standards of evidence; the AI collaborators carried out the mathematical development within the author's coordinate system — formulations, proofs, computations, and connections to the literature. Because this division of labor is unusual, correctness claims rest on the documented verification protocol rather than on authorial authority alone: every computational claim is backed by a runnable script, named here or at the record section cited; every result labeled proved was re-verified with independently written code before being so labeled; the calibration campaign's four dissolved anomalies, and two corrected search-filter errors on the cycle side, are documented rather than smoothed over. The complete record (wiki, code, draft history, origin note) is public at https://github.com/macindoe/collatz; every wiki section and script named in this paper is cited at commit [**record commit, pinned at release**].

B The reduced system: definitions and proofs

This appendix states formally what Sections 2 and 3 tell in words.

Definition B.1. *A* reduced state *is a pair* (ω, d) *with* ω *odd and positive,* $3 \nmid \omega$, $d \geq 1$. *Set*

$$A = 3^d\omega - 1, \qquad s = v_2(A)\,, \qquad x_{\text{exit}} = A/2^s, \qquad C = A + 2^s,$$

$$\sigma = v_2(C)\,, \qquad a_+ = v_3(C)\,, \qquad m_+ = \sigma - s, \qquad \omega_+ = \frac{C}{2^\sigma 3^{a_+}}, \qquad d_+ = m_+ + a_+,$$

and define the reduced map $F(\omega, d) = (\omega_+, d_+)$. *The* projection R *sends an odd* x *with* $x + 1 = 2^m 3^a \omega$ *(*$3 \nmid \omega$*) to* $R(x) = (\omega, m + a)$.

Proposition B.2 (block structure and well-definedness)**.** *Let* x *be odd with* $x + 1 = 2^m u$, $u = 3^a\omega$ *odd,* $3 \nmid \omega$, *and let* $(\omega, d) = R(x)$, $d = m + a$. *Then:*

(i) the next $m - 1$ *iterates of* T *are* x_j *with* $x_j + 1 = 2^{m-j}\,3^j u$ *for* $j = 0, \ldots, m - 1$;

(ii) the following iterate is $T(x_{m-1}) = A/2^s = x_{\text{exit}}(\omega, d)$, *which depends only on* (ω, d);

(iii) consequently $R(T^m(x)) = F(R(x))$ *for every representative* x *of* (ω, d), *the representatives being exactly* $x = 2^{d-a}3^a\omega - 1$, $0 \leq a \leq d - 1$; *and the sequence of reduced states visited by any* T*-orbit is the* F*-orbit of its projection.*

Proof. (i) If $y+1 = 2^k v$ with $k \geq 2$, v odd, then $3y+1 = 3(y+1)-2 = 2\left(3 \cdot 2^{k-1}v - 1\right)$, an odd multiple of 2, so $T(y) = 3 \cdot 2^{k-1}v - 1$ and $T(y)+1 = 2^{k-1}(3v)$: one factor of 2 is exchanged for one factor of 3. Induction from $y = x$ gives (i). (ii) At $j = m-1$ we have $x_{m-1}+1 = 2 \cdot 3^{m-1}u$, so $3x_{m-1}+1 = 3(x_{m-1}+1) - 2 = 2(3^m u - 1) = 2A$ since $3^m u = 3^{m+a}\omega = 3^d\omega$. Thus $T(x_{m-1}) = 2A/2^{1+s} = A/2^s$, and A depends only on (ω, d). (iii) $x_{\text{exit}}+1 = (A+2^s)/2^s = C/2^s$, so $R(x_{\text{exit}}) = (\omega_+, (\sigma - s) + a_+) = F(\omega, d)$ by Definition B.1; the representative list is immediate from $x+1 = 2^{d-a}3^a\omega$. □

Theorem B.3 (convergence translation)**.** $(1,1)$ *is a fixed point of* F *whose* R*-fiber is* $\{1\}$*. The Collatz conjecture holds if and only if every* F*-orbit from a valid state reaches* $(1,1)$*, and* R *induces a bijection between nontrivial* T*-cycles and* F*-cycles other than* $\{(1,1)\}$*.*

Proof. For $(1,1)$: $A = 2$, $s = 1$, $x_{\text{exit}} = 1$, $C = 4$, so $F(1,1) = (1,1)$; the fiber is $\{1\}$ since the representatives are $x = 2 \cdot 1 - 1 = 1$ only. If the conjecture holds and (ω, d) is valid, its representative $x = 2^d\omega - 1$ reaches 1, so by Proposition B.2(iii) the F-orbit of (ω, d) reaches $R(1) = (1,1)$. Conversely if every F-orbit reaches $(1,1)$, then any odd x has a T-orbit visiting the fiber of $(1,1)$, i.e. the value 1. For cycles: a nontrivial T-cycle projects to a periodic F-orbit avoiding $(1,1)$ (its values avoid 1); conversely an F-cycle's block entries $e_t = x_{\text{exit}}(S_{t-1})$ are deterministic in the states and joined by genuine T-iteration, so e_0 is T-periodic and nontrivial; the two constructions are mutually inverse. □

C The exact laws: statements and proofs

This appendix states formally what Section 4 tells in words.

For $u \equiv 1 \pmod 8$ the equation $9^t = u^{-1}$ has a unique solution $t \in \mathbb{Z}_2$, since 9 topologically generates $1 + 8\mathbb{Z}_2$; write $N(u)$ for it, so $N(u) = -\log u / \log 9$ in the 2-adic logarithm. N is an isomorphism $(1 + 8\mathbb{Z}_2, \times) \to (\mathbb{Z}_2, +)$ — the inverse of $t \mapsto 9^{-t}$, a continuous homomorphism, injective by (1) below and onto because its image is compact and dense — and

$$v_2\left(9^t - 1\right) = 3 + v_2(t) \qquad (t \in \mathbb{Z}_2 \setminus \{0\}), \tag{1}$$

the *isometry*, since $9 = 1 + 8$ and $v_2(\log(1+8z)) = 3 + v_2(z)$-type estimates hold with equality here.

Definition C.1 (anchor)**.** *For every odd* ω *set* $M(\omega) = N(\omega^2) = -2\log\omega/\log 9 \in \mathbb{Z}_2$*, well defined since* $\omega^2 \equiv 1 \pmod 8$*.* $M(\omega)$ *is even iff* $\omega \equiv \pm 1 \pmod 8$*.*

The parity clause: $\omega^2 \equiv 1 \pmod{16}$ for $\omega \equiv \pm 1 \pmod 8$ and $\omega^2 \equiv 9 \pmod{16}$ otherwise, while $9^t \equiv \omega^{-2} \pmod{16}$ and $9^2 \equiv 1 \pmod{16}$ force $t = M(\omega)$ even in the first case and odd in the second.

Lemma C.2 (squaring lemma)**.** *Let* ω *be odd,* $d \geq 1$*, and* $Y = 3^d\omega$*. If* $Y \equiv 1 \pmod 8$*, then*

$$v_2(Y-1) = 2 + v_2(d - M(\omega))\,.$$

Proof. $Y^2 = 9^d\omega^2 = 9^d\, 9^{-M(\omega)} \cdot \left(9^{M(\omega)}\omega^2\right) = 9^{\,d-M(\omega)}$, since $9^{M(\omega)} = \omega^{-2}$ by definition of N. By (1), $v_2\left(Y^2-1\right) = 3 + v_2(d - M(\omega))$ (nonzero since $Y^2 \neq 1$ for a valid state). Since $Y \equiv 1 \pmod 8$ we have $v_2(Y+1) = 1$, and $v_2(Y-1) = v_2\left(Y^2-1\right) - v_2(Y+1)$ gives the claim. □

Theorem C.3 (global valuation law)**.** *Call* (ω, d) lifting *if* $3^d\omega \equiv 1 \pmod 8$*, i.e.* $\omega \equiv 1\ (8)$ *with* d *even or* $\omega \equiv 3\ (8)$ *with* d *odd. On the lifting classes*

$$s = v_2\Big(3^d\omega - 1\Big) = 2 + v_2(d - M(\omega)) \ \geq\ 3,$$

while on the six non-lifting residue-parity classes $s \in \{1, 2\}$ *is the constant determined by* ($\omega \bmod 8$, $d \bmod 2$)*:* $s = 1$ *on* $(1, \text{odd}), (5, \text{odd}), (7, \text{even}), (3, \text{even})$ *and* $s = 2$ *on* $(5, \text{even}), (7, \text{odd})$*. Unconditionally* $s \leq C(\omega)\,(1 + \log d)^2$*: for* $\omega \neq 1$ *by p-adic Baker theory [22], [23, Corollaire 2], and at* $\omega = 1$ *exactly, since there the lifting class is* d *even and* $s = 2 + v_2(d)$*.*

Proof. The lifting case is Lemma C.2; the parity claim ($d \equiv M(\omega) \bmod 2$ on lifting classes) follows from $v_2(d - M(\omega)) \geq 1$ there. Off-lifting, $3^d\omega \bmod 8 \in \{3, 5, 7\}$ gives $v_2\big(3^d\omega - 1\big) \in \{1, 2\}$ by inspection of residues mod 8. The Baker bound applies to $s = v_2(9^n\tilde{\omega} - 1)$, a linear form in two 2-adic logarithms, for $\tilde{\omega} \neq 1$ (here $\tilde{\omega} \in \{\omega, 3\omega\}$ according to the lifting component, so $\tilde{\omega} = 1$ only at $\omega = 1$, d even); at $\tilde{\omega} = 1$ the pair $(9, 1)$ is multiplicatively dependent and the Baker input does not apply, but $M(1) = 0$ and the lifting case above gives $s = 2 + v_2(d) \leq 2 + \log_2 d$ exactly, by the isometry (1). □

Lemma C.4 (3-adic absorption law)**.** *Let* $h(s) = v_3(2^s - 1)$*, so* $h(s) = 0$ *for* s *odd and* $h(s) = 1 + v_3(s)$ *for* s *even. Then*

$$a_+ = v_3(C) = \begin{cases} 0, & s \text{ odd}, \\ \min(d,\ h(s)), & s \text{ even},\ d \neq h(s), \\ d + v_3\big(\omega + (2^s - 1)3^{-d}\big)\,, & s \text{ even},\ d = h(s). \end{cases}$$

In particular the next state gains a factor of 3 *exactly when* s *is even.*

Proof. $C = 3^d\omega + (2^s - 1)$ with $v_3\big(3^d\omega\big) = d$ exactly and $v_3(2^s - 1) = h(s)$; $h(s) = 0$ iff s odd since $2^s \equiv 2 \pmod 3$ then, and for even s, lifting-the-exponent gives $v_3\big(4^{s/2} - 1\big) = v_3(3) + v_3(s/2) = 1 + v_3(s)$. The ultrametric equality $v_3(X + Y) = \min(v_3(X)\,, v_3(Y))$ holds when the valuations differ; when $d = h(s)$, factor 3^d and evaluate the remaining unit sum. □

Theorem C.5 (depth closure)**.** *On every residue class the next depth is* $d_+ = m_+ + a_+$ *with* a_+ *given exactly by Lemma C.4 and* m_+ *by the* entry-depth law*: writing* ξ *for the anchor displacement* ($\xi = n - N(\omega)$ *on the lifting class* $d = 2n$*,* $\xi = n - N(3\omega)$ *on* $d = 2n + 1$*) and* $\delta_s = N(1 - 2^s)$ *(which satisfies* $v_2(\delta_s) = s - 3$ *for* $s \geq 3$*),*

$$m_+ = 3 - s + v_2(\xi + \delta_s) \quad \textit{(lifting)}, \qquad m_+ = \begin{cases} 1 + v_2(d - M(\omega))\,, & s = 1, \\ v_2((d - 1) - M(\omega))\,, & s = 2. \end{cases}$$

The forced carry $v_2(\delta_s) = s - 3 = v_2(\xi)$ *yields* $m_+ \geq 1$ *structurally.*

Proof. All cases reduce to the *target-shift lemma*: for $c \in 1 + 8\mathbb{Z}_2$ and $d = 2n$ with $\omega \equiv 1\ (8)$,

$$v_2\Big(3^d\omega - c\Big) \ =\ 3 + v_2(n - N(\omega) + N(c))\,,$$

proved exactly as Lemma C.2 after writing $3^d\omega - c = 9^{-N(c)}\big(9^{\,n - N(\omega) + N(c)} - 1\big)$. Since $m_+ = v_2(x_{\text{exit}} + 1)$ and $2^s(x_{\text{exit}} + 1) = C = 3^d\omega - (1 - 2^s)$, the lifting case is the shift $c = 1 - 2^s$ (which

lies in $1 + 8\mathbb{Z}_2$ for $s \geq 3$), with $v_2(\delta_s) = s - 3$ from the isometry; the odd lifting component reduces to the even one via the companion parameter 3ω. For $s = 1$: the shifted target is -1, and $v_2(X+1) = v_2\big(X^2-1\big) - v_2(X-1)$ with $X = 3^d\omega$, $X^2 = 9^{\,d-M(\omega)}$, gives the formula; For $s = 2$: here $C = 3^d\omega + 3 = 3(X'+1)$ with $X' = 3^{d-1}\omega$, so $m_+ = v_2(C) - 2 = v_2(X'+1) - 2$. The $s = 2$ classes force $3^d\omega \equiv 5 \pmod 8$, hence $X' \equiv 3^{-1} \cdot 5 \equiv 7 \pmod 8$, so $v_2(X'-1) = 1$; and $X'^2 = 9^{\,(d-1)-M(\omega)}$ as in Lemma C.2, so $v_2(X'+1) = v_2\big(X'^2-1\big) - v_2(X'-1) = \big(3 + v_2((d-1) - M(\omega))\big) - 1$, giving $m_+ = v_2((d-1) - M(\omega))$ as claimed. □

Remark C.6 (verification notes). Each law above was checked by independently written code, with zero discrepancies: the valuation law on 8,000 random states across both lifting components, the entry-depth law exhaustively on all 62,937 valid states with $\omega < 3000$, $d < 64$, and the absorption law on 327,980 states (983,954 comparisons against both this Lemma and its form in the record), with the boundary branch $d = h(s)$, on which a_+ is a 3-adic function of ω, and the shallow branch $d < h(s)$ constructed rather than sampled. The records are `stage3.md` §11.8.6.3 and that page's status header; the code is `experiments/absorption_law.py`.

Theorem C.7 (low-order anchor-increment law)**.** *Let $\Delta M = M(\omega_+) - M(\omega)$. For every $k \geq 1$, $\Delta M \bmod 2^k$ is determined by, and explicitly computable from,*

$$\omega \bmod 2^{\sigma+k+2}, \qquad d \bmod 2^{\sigma+k}, \qquad a_+ \bmod 2^k,$$

and the stratum data (s, σ) are determined by the same residues. This last clause does not extend to a_+: by Lemma C.4 the absorption is a 3-adic function of ω, and it enters the display only through its residue $a_+ \bmod 2^k$.

Proof. Three digit-determinacy facts chain together. (a) For $u \equiv 1$ (8), $N(u) \bmod 2^k$ depends only on $u \bmod 2^{k+3}$, since 9 has order $2^k \bmod 2^{k+3}$ and $9^t \equiv u^{-1}$ determines $t \bmod 2^k$. (b) $C \bmod 2^q$ is determined by $\omega \bmod 2^q$, $d \bmod 2^{q-2}$, s, since 3 has order $2^{q-2} \bmod 2^q$. (c) $\omega_+ \bmod 2^r$ is determined by $C \bmod 2^{\sigma+r}$, σ, and $a_+ \bmod 2^{r-2}$, by exact division and the order of 3 mod 2^r. Now $\Delta M = N\big((\omega_+/\omega)^2\big)$; by (a) this needs $(\omega_+/\omega)^2 \bmod 2^{k+3}$, hence $\omega_+, \omega \bmod 2^{k+2}$; by (c) with $r = k+2$ this needs $C \bmod 2^{\sigma+k+2}$ and $a_+ \bmod 2^k$; by (b) this needs the stated residues of ω and d. Finally $s \leq \sigma$ and σ are visible in $A \bmod 2^{\sigma+1}$, $C \bmod 2^{\sigma+1}$. □

Theorem C.8 (one-step trichotomy)**.** *The* depth-k window *of a valid state (ω, d) consists of the residues of Theorem C.7 together with the stratum labels (s, σ, a_+). From the depth-k window alone, one can determine: the exact next depth d_+ and next residue-parity class; the exact value $s_+ \in \{1, 2\}$ if the next class is non-lifting; and, if lifting, the truncation of the next displacement mod 2^k — yielding s_+ exactly when nonzero, and the correct report "$s_+ \geq k+2$" when zero. The window never outputs a wrong value.*

Proof. By facts (b), (c) above the window determines $\omega_+ \bmod 2^{k+2}$, hence the next class; $d_+ = (\sigma - s) + a_+$ is exact from the stratum data. Non-lifting s_+ is the class constant (Theorem C.3). Lifting: $M(\omega_+) \bmod 2^k$ follows from fact (a) applied to ω_+^2, so $(d_+ - M(\omega_+)) \bmod 2^k$ is computable; if nonzero its valuation is visible and Theorem C.3 gives s_+; if zero then $v_2(d_+ - M(\omega_+)) \geq k$, i.e. $s_+ \geq k+2$. □

By the proof, the window is undecided exactly when $s_+ \geq k+2$. Under the law $\pi_{k,D}$ of Appendix E with $D \geq k+2$, whose s-marginal is $P(s = j) = 2^{-j}$ for $j < D$, that event has probability $\sum_{j \geq k+2} 2^{-j} = 2^{-(k+1)}$ exactly. Along real orbits the record measures an undecided rate of 0.0275 against 0.03125 at $k = 4$, and 0.0019 against 0.00195 at $k = 8$ (`stage4.md`, Remark "undecided rate").

Heuristic C.9 (digit budget). *Deciding one step at depth k consumes the state's 2-adic data to depth $\sigma+k+2$ ($\sigma = s+m_+$ has mean 4 under the model law of Appendix E, where s and m_+ are geometric of mean 2; the heuristic does not depend on the constant). Theorem C.7 shows that these residues suffice; that they are also needed is part of the heuristic. Nothing regenerates them: the anchors of later states are functions of the full initial state, not of any bounded window. A window of W digits therefore supports $O(W/(k+6))$ decided steps. This strongly suggests — and we treat it as the organizing heuristic, not a formalized theorem — that bounded-window determinism cannot decide unbounded horizons without additional statistical or rigidity input.*

Remark C.10 (verification note). The trichotomy was run along real orbits for $k \in \{4, 8\}$: 21,296 steps, 21,000 decided with zero errors, 296 flagged deep with zero violations of the bound $s_+ \geq k+2$. Code: `experiments/one_step_propagation.py`; the low-order law of Theorem C.7 is checked by `experiments/anchor_increment.py`.

D Cycles: the apparatus and the proofs

This appendix states formally what Section 5 tells in words.

Fix a hypothetical p-block cycle of F with entry depths $m_t \geq 1$, exit valuations $s_t \geq 1$ ($t \in \mathbb{Z}/p\mathbb{Z}$), and set $n = \sum m_t$, $K = \sum s_t + n$, $q = 2^K - 3^n$.

Notation. Two letters keep two meanings, to match the record or the literature: M_t (Proposition D.1) is a partial sum of entry depths, unrelated to the anchor $M(\omega)$; and S_t (Proposition D.1) and S_n (Hypothesis E.1) are both partial sums of exponents of 2, over different ranges. Λ is the linear form $K\log 2 - n\log 3$. Symbols that carried two meanings in v3 are renamed in this version; Appendix H lists them.

Proposition D.1 (elimination identity). *$q > 0$, and for every rotation r,*

$$\omega_r\, 3^{a_{r-1}}\, q \;=\; R_r \;:=\; \sum_{t=0}^{p-1} 3^{M_t}\, 2^{S_t}\, (2^{s_t} - 1),$$

where, in rotation order starting at r, $M_t = \sum_{j>t} m_j$ and $S_t = \sum_{j<t} \sigma_j$, with $\sigma_j = s_j + m_{j+1}$, indices cyclic (the step's $\sigma = v_2(C)$, $m_+ = \sigma - s$, of Definition B.1; the shift is essential, m_{j+1} and not m_j; and M_t, a partial sum of entry depths, is unrelated to the anchor $M(\omega)$ of Appendix C — this is the project record's own notation, `cycles.md` §12.6.1, kept so that paper and record read together). Hence $q \mid R_r$ and $q \leq \min_r R_r$. For the trivial cycle, $R = 4^p - 3^p = q$ identically.

This is the classical cycle equation of Böhm and Sontacchi [11] (see [21, I, item 26]) in block coordinates; in integer form it reads $2^{m_0} R_0 = 2^K F_n(a) + q$, with F_n Tao's n-Syracuse offset [4, (1.5)] and a the valuation word of the profile (`cycles.md` Remark 12.6.1.7).

Proof. The step identity $2^{\sigma_t} 3^{a_t} \omega_{t+1} = 3^{d_t}\omega_t + (2^{s_t} - 1)$ (Definition B.1) unrolls p times; imposing closure $\omega_p = \omega_0$, substituting $d_j = m_j + a_{j-1}$, and cancelling the common power of 3 yields the display. Positivity: the right side is positive. □

Corollary D.2 (size balance). *$2^K = 3^n \prod_t (1 + \chi_t)$ with $\chi_t = (2^{s_t} - 1)/(3^{d_t}\omega_t)$; if every cycle element exceeds X then $0 < K\log 2 - n\log 3 < p/X$. This is the classical Baker-side engine [18], obtained in four lines.*

Proof. Multiplying the step identity $2^{\sigma_t} 3^{a_t} \omega_{t+1} = 3^{d_t} \omega_t (1+\chi_t)$ around the cycle, the ω-products cancel, and $\sum_t \sigma_t = K$ and $\sum_t d_t = n + \sum_t a_t$ leave $2^K = 3^n \prod_t (1+\chi_t)$. Each $\chi_t > 0$, so $K \log 2 - n \log 3 = \sum_t \log(1+\chi_t)$ lies in $(0, \sum_t \chi_t)$. Finally $3^{d_t}\omega_t > A_t = 2^{s_t} x_t$, where x_t, the exit of block t, is a cycle element, so $\chi_t < 1/x_t < 1/X$. (`cycles.md` Corollary 12.1.2.) □

Lemma D.3 (ceiling)**.** *If $K > \lceil n \log_2 3 \rceil$ then $\prod(1+\chi_t) \geq 2$, so some $\chi_t \geq 2^{1/p} - 1 \geq (\ln 2)/p$ and some block exit is $< p/\ln 2$. Consequently $K = \lceil n \log_2 3 \rceil$ for every nontrivial cycle with $p \leq \ln 2 \cdot X_0$, X_0 the verified range [25] — unconditionally for $p \leq 5$ by inspecting the odd values ≤ 7.*

Proof. $K \geq \lceil n \log_2 3 \rceil + 1 > n \log_2 3 + 1$ gives $\prod_t (1+\chi_t) = 2^K/3^n > 2$ (Corollary D.2), so some $1 + \chi_t \geq 2^{1/p}$, and $\chi_t \geq 2^{1/p} - 1 \geq (\ln 2)/p$; since $\chi_t < 1/x_t$ (proof of Corollary D.2), the exit x_t of that block is $< p/\ln 2$. As $q > 0$ forces $K \geq \lceil n \log_2 3 \rceil$, a nontrivial cycle with $K \neq \lceil n \log_2 3 \rceil$ therefore has an element below $p/\ln 2 \leq X_0$, which reaches 1: impossible. For $p \leq 5$, $p/\ln 2 < 7.22$, and the odd values $1, 3, 5, 7$ all reach 1 ($3 \to 5 \to 1$; $7 \to 11 \to 17 \to 13 \to 5 \to 1$), so no nontrivial cycle contains them. (`cycles.md` Lemma 12.6.2 and Corollary 12.6.3.) □

Theorem D.4 (periods 1–3; rederivations)**.** *F has no nontrivial cycles of 1, 2, or 3 blocks. These statements are contained in the classical results [8, 9, 19, 20]; the derivations, not the theorems, are the contribution.*

Proof outline. $p = 1$: the fixed-point equation is $\omega\, 3^a (2^{s+m} - 3^m) = 2^s - 1$; the sign and size constraints pin 2^s into a window of endpoint ratio $(1 - 3^{-m})/(1 - 2^{-m}) < 2$, leaving at most one candidate s per m; exact search kills $m \leq 20{,}000$ and Steiner's theorem [8] covers all m via the block–circuit correspondence. $p = 2, 3$: budget-trim lemmas (casework over rotations' dominant terms, using $\sum s_t = K - n$) force $0 < K \log 2 - n \log 3 < 2^{6-n/5}$ resp. $2^{3-0.115n}$; exact searches to $n = 20{,}000$ find no solutions; any effective irrationality measure for $\log_2 3$ [24] excludes the tail. Full details are in the project record: the three classifications at `cycles.md` §12.2.3, §12.5.3 and §12.7.5, the two trim lemmas at §12.5.2 and §12.7.4, and the ceiling lemma that supplies $K = \lceil n \log_2 3 \rceil$ at §12.6.2. The searches are `experiments/period1_cycles.py`, `experiments/period2_cycles.py` and `experiments/period3_cycles.py`. □

The dictionary with m-cycles. In the Terras map $x \mapsto (3x+1)/2$ (x odd), $x \mapsto x/2$ (x even), a block of entry depth m_t and exit valuation s_t is exactly m_t odd terms followed by s_t even terms: by Proposition B.2(i)–(ii) each of $x_0, \ldots, x_{m-1}$ is odd and maps to the next by $(3x+1)/2$, the last of them to the even number $A = 3^d \omega - 1$, which halves $s = v_2(A) \geq 1$ times down to the next block entry. So a p-block cycle of F has exactly p maximal runs of odd terms, and its local minima are exactly its p block entries, each lying below both its predecessor $2x$ and its successor $(3x+1)/2$ — including the entries of depth $m_t = 1$, which look like pure falls only in the odd-to-odd orbit. Hence a p-block cycle is an m-cycle of Simons and de Weger [19] with $m = p$, their k_i our m_t, their ℓ_i our s_t, their K our n and their $K + L$ our K; Rozier's number of local minima [10] is the same count. Every m-cycle exclusion therefore transfers verbatim: Hercher's theorem [20] excludes every nontrivial cycle of F with $p \leq 91$ blocks, of which Theorem D.4 is a small special case. Such exclusions bound the number of blocks only, and say nothing about how many blocks have depth $m_t \geq 2$ (`cycles.md` §12.4).

Configurations. The proofs below use a cycle only through its block data. A *p-block configuration* is a tuple of integers $m_t, s_t \geq 1$ ($t \in \mathbb{Z}/p\mathbb{Z}$) with n, K, q as above and $q > 0$; its

rotation numerators R_r are defined by the display of Proposition D.1, and $\gamma := K - \log_2 q$. It is a *size-passer* if $q \le R_r$ for every r. Put $x_r := 2^{m_r} R_r/q - 1$ and $x_{\min} := \min_r x_r$. On a cycle, Proposition D.1 gives $x_r + 1 = 2^{m_r} 3^{a_{r-1}} \omega_r$, so x_r is the entry of block r and, by the dictionary above, $x_{\min}$ is the least element of the cycle; for a configuration the same unrolling over the rationals makes the x_r the block entries of the rational cycle it defines. Write $\Lambda := K \log 2 - n \log 3$; since $3^n/2^K = 1 - q/2^K = 1 - 2^{-\gamma}$, we have $\Lambda = -\log(1 - 2^{-\gamma}) > 2^{-\gamma}$. Write $\kappa_p := (\log_2 3 - 1)/((\log_2 3)^p - 1)$ for the trim's exponent (the record's ε_p) and, for an arc A of consecutive blocks, $w(A) := (\log_2 3 - 1)\, m(A) - s(A)$, its m-mass at exchange rate $\log_2 3 - 1$ minus its s-mass. For each rotation r and each position t in rotation order from r, let A_t be the arc of the blocks after position t (it ends at block $r-1$, and is empty for $t = p-1$). Then the terms of R_r are

$$3^{M_t}\, 2^{S_t}\, (2^{s_t} - 1) \;=\; 2^{K - m_r + w(A_t)}\, \big(1 - 2^{-s_t}\big), \tag{2}$$

since $M_t = m(A_t)$ and $S_t + s_t = (K - n - s(A_t)) + (n - m_r - M_t)$.

Theorem D.5 (uniform trim; Rozier [10], Simons–de Weger [19])**.** *Every nontrivial p-block cycle satisfies, with $\gamma := K - \log_2 q$,*

$$\gamma + \log_2 p \;>\; \frac{(\log_2 3 - 1)\, n}{(\log_2 3)^p - 1}.$$

Consequently the candidate set at every period is finite and explicitly bounded. Combined with Rhin's effective bound [24],

$$K \log 2 - n \log 3 \;>\; \exp\big(-13.3\,(0.46057 + \log n)\big),$$

every nontrivial p-block cycle has $n \le n_0(p)$, the unique solution in n of

$$\frac{(\log_2 3 - 1)\, n}{(\log_2 3)^p - 1} \;=\; \log_2 p \;+\; \frac{p + 13.3\,(0.46057 + \log n)}{\log 2},$$

so that $n_0(p) = O(p\,(\log_2 3)^p)$.

Proof. For each rotation, $q \le R_r < p\, 2^{\max_t e_t}$, where the exact exponent identity gives $e_t - K = (\log_2 3 - 1) M_t - m_r - \sum_{j>t} s_j$. The resulting conditions $m_r < \gamma + \log_2 p + \Phi_r$ close into the cyclic recursion $\Phi_r = \max(0, \Phi_{r-1} + w_{r-1})$ on arc weights $w_j = (\log_2 3 - 1) m_j - s_j$, whose cyclic total $(\log_2 3 - 1) n - \sum s_t$ is negative because $2^K > 3^n$ forces $\sum s_t > (\log_2 3 - 1) n$. A max-plus recursion with negative cyclic total must vanish somewhere (otherwise following it once around the cycle would add the total to itself); fix such a reset index r_0, so $\Phi_{r_0} = 0$. Walking forward, $\Phi_{r_0 + j} \le (\log_2 3 - 1)\mu_{j-1}$ where μ_j is the partial sum $m_{r_0} + \cdots + m_{r_0 + j}$ (drop the $-s$ terms), so the block bound gives $m_{r_0+j} < (\gamma + \log_2 p) + (\log_2 3 - 1)\mu_{j-1}$, i.e. $\mu_j < \log_2 3\, \mu_{j-1} + (\gamma + \log_2 p)$. Iterating p times from $\mu_{-1} = 0$ yields $n = \mu_{p-1} < (\gamma + \log_2 p)\, \big(\log_2 3^p - 1\big)/(\log_2 3 - 1)$, which is the display. The argument uses only $q \le R_r$ for every r, so the first display holds for every size-passing configuration. □

This is Rozier's Lemme 4 [10, p. 11, with (5) on p. 10] and Simons and de Weger's Lemma 7 [19, p. 58] in rotation coordinates, up to an additive constant, with their number of local minima equal to our p by the dictionary above; finiteness at each p is due to Brox [13]. The second display is [19, Lemma 14 and Theorem 3(d)], whose $K_1(p)/(p\,(\log_2 3)^p)$ decreases to $15.107\ldots$, with the Rhin measure used there [19, Lemma 12]; its derivation from the first display and

Rhin's bound, in the present notation, is `cycles.md` Corollary 12.8.2. We rederive it by another route — rotation numerators and a max-plus recursion, where Simons and de Weger chain the minima — because the recursion identifies the extremal profile that Theorem D.6 realizes.

Theorem D.6 (sharpness: the staircase)**.** *Write $\phi(x) := -\log_2(1-2^{-x})$ for $x > 0$, a decreasing involution of $(0,\infty)$.*

(i) Every period. *For every $p \geq 16$ there are an integer $n \in [(\log_2 3)^p,\ 1.05\,(\log_2 3)^p]$ and a p-block size-passer with $\sum_t m_t = n$, $K = \lceil n \log_2 3 \rceil$ and*

$$3.683012 \ \leq\ \gamma \ \leq\ 5.140212.$$

Size-passers in the same bracket exist at every period $3 \leq p \leq 15$, and size-passers exist at $p \in \{2,4\}$. Consequently, if $\gamma + \log_2 p > c_p\, n$ for every size-passer at a period $p \geq 16$, then $c_p < (5.140212 + \log_2 p)\,(\log_2 3)^{-p}$: the exponential loss in Theorem D.5 is intrinsic to size arguments.

(ii) Necessity at a size threshold. *Every configuration satisfies*

$$\gamma - 1 \ \leq\ \log_2(x_{\min} + 1) \ <\ \gamma + \log_2 p.$$

Hence for real $X_0 \geq 1$, every size-passer with $x_{\min} \geq X_0$ has $\gamma > \max\big(\kappa_p n,\ \log_2(X_0 + 1)\big) - \log_2 p$, and every configuration with $\gamma \geq \log_2(X_0+1) + 1$ has $x_{\min} \geq X_0$.

(iii) Sufficiency at a size threshold. *Let $X_0 \geq 1$ be real, $\lambda_0 := \log_2 3/(\log_2 3 - 1) + 2^{-20} = 2.7095122\ldots$ and $g := 1 + \max\big(\log_2(X_0+1),\ \lambda_0\big)$. For every period $p \geq p_1(X_0)$, with $p_1(X_0)$ defined in the proof from the continued fraction of $\log_2 3$, there is a p-block size-passer with $x_{\min} \geq X_0$ and*

$$g \leq \gamma \leq g + 1, \qquad \lambda_0 \ \leq\ \gamma - \kappa_p n \ \leq\ \lambda_0 + 1.5, \qquad \kappa_p n \ \leq\ \rho_0 \ \leq\ \gamma,$$

where $\rho_0 := \max_j\big(m_j - (\log_2 3 - 1)(m_0 + \cdots + m_{j-1})\big)$ is Rozier's quantity in block order from block 0. This configuration has $\kappa_p n > g - \lambda_0 - \frac{1}{2}$ and $\Lambda > 2^{-\gamma} > 0.054 \cdot 2^{-\kappa_p n}$.

(iv) Instances. *At $p = 7$, $n = 94$, the staircase $m = (4,7,9,15,23,35,1)$ with $s = (1,\ldots,1,\ s_{\text{tot}} - 6)$ passes all seven size tests with $\gamma = 6.74$, where the period-3 constant would demand $\gamma > 8.9$, and 84 further instances occur at $p = 6$ alone; these fail the divisibility conditions $q \mid R_r$.*

Uniform cycle exclusion therefore requires the divisibility conditions $q \mid R_r$ of Proposition D.1, not size alone — the conclusion Halbeisen and Hungerbühler reached in the cycle-length parameter [12, p. 238].

Proof. The function ϕ. $2^{-x} + 2^{-\phi(x)} = 1$ is symmetric in x and $\phi(x)$, so ϕ is an involution; it decreases because $1 - 2^{-x}$ increases.

Two facts used throughout. (a) For $K = \lceil n \log_2 3 \rceil$ put $\delta_n := K - n\log_2 3 \in (0,1)$; then $q/2^K = 1 - 2^{-\delta_n}$, so $\gamma = \phi(\delta_n)$, and $\delta_{n+j} \equiv \delta_n - j \log_2 3 \pmod 1$. (b) Multiplying (2) by $2^{m_r}/q = 2^{m_r + \gamma - K}$ and summing over t gives, for every rotation r,

$$x_r + 1 \ =\ 2^{\gamma} \sum_{t=0}^{p-1} 2^{w(A_t)} \big(1 - 2^{-s_t}\big). \tag{3}$$

(ii). In (3) the empty arc ($t = p-1$) contributes $1-2^{-s_{r-1}} \geq \frac{1}{2}$ and every term is positive, so $x_r + 1 \geq 2^{\gamma-1}$ for every r. For the upper bound put $W_j := \sum_{i<j}\big((\log_2 3-1)m_i - s_i\big)$, so $W_0 = 0$ and $W_p = n\log_2 3 - K < 0$; choose $j^* \in \{1,\dots,p\}$ minimizing W_j and put $r_0 := j^* \bmod p$. An arc ending at block $r_0 - 1$ either does not wrap, with weight $W_{j^*} - W_i \leq 0$ ($0 \leq i \leq j^*$; at $i = 0$ the weight is $W_{j^*} \leq W_p < 0$), or wraps, with weight $W_p + W_{j^*} - W_i < 0$; so every term of (3) at $r = r_0$ is < 1, and $x_{r_0} + 1 < p\,2^{\gamma}$. With Theorem D.5, whose proof uses only the size conditions, a size-passer with $x_{\min} \geq X_0$ has $\gamma + \log_2 p > \kappa_p n$ and $\gamma + \log_2 p > \log_2(x_{\min}+1) \geq \log_2(X_0+1)$; and $\gamma \geq \log_2(X_0+1)+1$ gives $x_{\min} + 1 \geq 2^{\gamma-1} \geq X_0 + 1$.

The construction. Given $p \geq 2$ and $n \geq 1$, let $K = \lceil n\log_2 3\rceil$, $s_{\mathrm{tot}} = K - n$, $q = 2^K - 3^n$, $\gamma = \phi(\delta_n)$, $s_c := s_{\mathrm{tot}} - p + 1$ and $\eta := \phi(s_c)$, and assume

$$\text{(H0)}\quad s_{\mathrm{tot}} \geq p \ \text{ and } \ (\log_2 3-1)(n-p) + \gamma \geq p + \eta; \qquad \text{(H1)}\quad \gamma \geq \max\big(2+\eta,\ \Gamma'(p,n)\big),$$

where

$$\Gamma'(p,n) := 1 + \eta + \frac{(\log_2 3)^p - (\log_2 3)^2 - (\log_2 3-1)(p-2) + (\log_2 3-1)^2 n}{(\log_2 3-1)\big((\log_2 3)^p - 1\big)} \;=\; \kappa_p n + \Gamma'_0(p) + \eta,$$

with $\Gamma'_0(p) < \log_2 3/(\log_2 3 - 1)$, the last inequality being $(\log_2 3)^2 + (\log_2 3-1)(p-2) > 1$. Set $s_t := 1$ for $t \leq p-2$ and $s_{p-1} := s_c$, and for $r = 0,\dots,p-2$, with $\mu_{-1} := 0$,

$$f_r := \gamma + (\log_2 3 - 1)\,\mu_{r-1} - r - \eta, \qquad b_r := n - 1 - \mu_{r-1} - (p-2-r),$$
$$m_r := \min\big(\lfloor f_r\rfloor,\, b_r\big), \qquad \mu_r := \mu_{r-1} + m_r,$$

and $m_{p-1} := n - \mu_{p-2}$, the *crash* block. The floor is an exact integer comparison: $\lfloor f_r\rfloor = \max\{m : q\,2^m \leq X_r\}$ with $X_r := 3^{\mu_{r-1}}\,2^{\,n-\mu_{r-1}+p-1-r}\,(2^{s_c}-1)$, since $\log_2 X_r = K - r - \eta + (\log_2 3 - 1)\mu_{r-1}$. We show that every $m_t \geq 1$, that $\sum_t m_t = n$ and $n + \sum_t s_t = K$, that the configuration is a size-passer, and that $\kappa_p n \leq \rho_0 \leq \gamma - \eta$.

Well-definedness. From $p \leq s_{\mathrm{tot}} < (\log_2 3 - 1)n + 1$ we get $n > (p-1)/(\log_2 3 - 1)$, hence $n \geq p$ and $b_0 = n - p + 1 \geq 1$; and $b_{r+1} = b_r - m_r + 1 \geq 1$ since $m_r \leq b_r$. So every climb block has $m_r \geq 1$ as soon as $\lfloor f_r\rfloor \geq 1$, and $\mu_{p-2} \leq n-1$, so the crash has $m_{p-1} \geq 1$. While the budget b_r has not bound, $m_r = \lfloor f_r\rfloor$: here $f_0 = \gamma - \eta \geq 2$, and $f_{r+1} - f_r = (\log_2 3 - 1)m_r - 1 > 0$ whenever $m_r \geq 2$, so $\lfloor f_r\rfloor \geq 2$ throughout. Once the budget binds at some r, every later b_j equals 1, $\mu_{j-1} = n - p + j$, and $f_j = \big(\gamma + (\log_2 3-1)(n-p) - \eta\big) - (2 - \log_2 3)\,j \geq p - (2-\log_2 3)(p-2) \geq 2$ by (H0), so $m_j = 1$.

The climb rotations $r \leq p-2$. In rotation order from r the crash block sits at position $t = p-1-r$, with $M_t = \mu_{r-1}$ and $S_t = (p-1-r) + (n-\mu_r)$, so its term of R_r is $3^{\mu_{r-1}}2^{\,n-\mu_r+p-1-r}(2^{s_c} - 1) = X_r/2^{m_r} \geq q$, because $m_r \leq \lfloor f_r\rfloor$. Every term is positive, so $q \leq R_r$.

The crash rotation $r = p-1$. Its position-0 term is $3^{\mu_{p-2}}(2^{s_c} - 1)$, and this is $\geq q$ if and only if

$$\text{(D)} \qquad \log_2 3\,\mu_{p-2} \;\geq\; n + p - 1 - \gamma + \eta.$$

If the budget binds somewhere, $\mu_{p-2} = n-1$ and (D) reads $(\log_2 3 - 1)n \geq \log_2 3 + p - 1 - \gamma + \eta$, which follows from (H0) because $(\log_2 3 - 1)(p-1) \geq 0$. If it never binds, $m_r = \lfloor f_r\rfloor > f_r - 1$ gives $\mu_r > \log_2 3\,\mu_{r-1} + \gamma - r - \beta$ with $\beta := 1 + \eta$, so by induction $\mu_r > A_r$ for the real recursion $A_r = \log_2 3\,A_{r-1} + \gamma - r - \beta$, $A_{-1} = 0$, whose solution is

$$A_r \;=\; \frac{(\log_2 3)^{r+1}\big((\gamma-\beta)(\log_2 3 - 1) - 1\big) + (r + \beta - \gamma)(\log_2 3 - 1) + \log_2 3}{(\log_2 3 - 1)^2}.$$

Clearing denominators, $\log_2 3\, A_{p-2} - (n+p-1-\gamma+\eta) = \big(((\log_2 3)^p - 1)/(\log_2 3 - 1)\big)\,\big(\gamma - \Gamma'(p,n)\big) \geq 0$ by (H1), which gives (D). (The crash depth is 1 exactly when the budget binds.)

Rozier's quantity. The climb terms satisfy $m_r - (\log_2 3 - 1)\mu_{r-1} \leq f_r - (\log_2 3 - 1)\mu_{r-1} = \gamma - r - \eta$, and the crash term is $n - \log_2 3\, \mu_{p-2} \leq \gamma + 1 - p - \eta$ by (D), so $\rho_0 \leq \gamma - \eta$. Conversely, all terms $\leq \rho_0$ give $\mu_j \leq \log_2 3\, \mu_{j-1} + \rho_0$ and $n \leq \rho_0((\log_2 3)^p - 1)/(\log_2 3 - 1)$, i.e. $\rho_0 \geq \kappa_p n$; this is Rozier's minimax [10, p. 11], valid for positive reals in any cyclic order.

(i). Let $\vartheta := 8 - 5\log_2 3 = 0.0751874964\ldots$, $\Gamma^* := 1.05\, \log_2 3(\log_2 3 - 1) + 1/(\log_2 3 - 1) + 1 = 3.6830121\ldots$, $\delta_{\mathrm{hi}} := \phi(\Gamma^*) = 0.1169390665\ldots$ and $\delta_{\mathrm{lo}} := 0.0415$. Since $5\log_2 3 = 8 - \vartheta$, $\delta_{n+5} \equiv \delta_n + \vartheta \pmod 1$. The fourteen points $j\vartheta \bmod 1$, $0 \leq j \leq 13$, are already sorted ($13\vartheta < 1$), with gaps ϑ and $1 - 13\vartheta < \vartheta$; so the fourteen closed arcs $[\delta_{\mathrm{lo}}, \delta_{\mathrm{hi}}] - j\vartheta$, of length $0.0754390665\ldots > \vartheta$, have left endpoints at mutual gaps at most ϑ and cover the circle. Hence for every N one of $\delta_N, \delta_{N+5}, \ldots, \delta_{N+65}$ lies in $[\delta_{\mathrm{lo}}, \delta_{\mathrm{hi}}]$: any 66 consecutive integers contain an n with $\delta_{\mathrm{lo}} \leq \delta_n \leq \delta_{\mathrm{hi}}$. For $p \geq 16$ the window $[(\log_2 3)^p, 1.05\,(\log_2 3)^p]$ contains at least $0.05\,(\log_2 3)^p - 1 > 78$ consecutive integers, hence such an n, and since ϕ decreases, $\Gamma^* \leq \gamma = \phi(\delta_n) \leq \phi(\delta_{\mathrm{lo}}) = 5.1402114\ldots$. For this n, $s_{\mathrm{tot}} > (\log_2 3 - 1)n \geq (\log_2 3 - 1)(\log_2 3)^p > p + 900$, so $\eta < 2^{-900}$, and $(\log_2 3 - 1)(n - p) > p + 1$, which gives (H0); and $\Gamma'(p,n) < 1.05\,(\log_2 3)^p\kappa_p + \log_2 3/(\log_2 3 - 1) + \eta < 3.33 < \Gamma^* \leq \gamma$ gives (H1). The construction applies. The periods $3 \leq p \leq 15$, at exhibited n with $n/(\log_2 3)^p \leq 1.70$, and the exhibitions at $p \in \{2, 4\}$, which this argument does not reach, are finite computations, recorded at `cycles.md` §12.8.6.1 and §12.8.6.4. Finally $n \geq (\log_2 3)^p$ and $\gamma \leq 5.140212$ give the bound on c_p.

(iii). Let P_j/Q_j be the convergents of $\log_2 3$, $\Theta_j := |Q_j \log_2 3 - P_j|$ and $\Delta := \phi(g) - \phi(g+1)$; let $N_* := Q_k + Q_{k-1}$ for the least $k \geq 1$ with $\Theta_{k-1} \leq \Delta$, and let $p_1(X_0)$ be the least p with $1/(2\kappa_p) = ((\log_2 3)^p - 1)/(2(\log_2 3 - 1)) \geq \max(N_*, 3p + 40)$ (the condition then holds at every larger p). Fix $p \geq p_1(X_0)$, put $n^* := \lfloor (g - \lambda_0)/\kappa_p \rfloor$ and take the window $[n^* - N_* + 1, n^*]$ of N_* consecutive integers. By (a), the points δ_n on it form a reflected translate of $\{j \log_2 3 \bmod 1 : 0 \leq j < N_*\}$, whose gaps, by the three-distance theorem in its two-length case [16, 17], are Θ_{k-1} and Θ_k, both at most Δ. So some n in the window has $\delta_n \in [\phi(g+1), \phi(g)]$, i.e. $g \leq \gamma \leq g+1$. On the window $n > (g - \lambda_0)/\kappa_p - N_* \geq 1/(2\kappa_p) \geq 3p + 40$, since $g - \lambda_0 \geq 1$; hence $s_{\mathrm{tot}} > (\log_2 3 - 1)n$ gives $s_c \geq 21$ and $\eta < 2^{-20}$, and $(\log_2 3 - 1)(n - p) \geq p + 1$, so (H0) holds. Also $\kappa_p n \leq \kappa_p n^* \leq g - \lambda_0$, so $\Gamma'(p,n) < \kappa_p n + \lambda_0 \leq g \leq \gamma$, and $\gamma \geq g \geq 1 + \lambda_0 > 2 + \eta$: (H1) holds, and the construction gives a size-passer. Since $\gamma \geq g \geq \log_2(X_0 + 1) + 1$, (ii) gives $x_{\min} \geq X_0$. For the brackets, $\gamma - \kappa_p n \geq g - (g - \lambda_0) = \lambda_0$; and $\kappa_p n > g - \lambda_0 - \kappa_p N_* \geq g - \lambda_0 - \frac{1}{2}$, whence $\gamma - \kappa_p n < \lambda_0 + \frac{3}{2}$. Finally $\Lambda > 2^{-\gamma} \geq 2^{-(\lambda_0 + 1.5)}\, 2^{-\kappa_p n}$, and $2^{-(\lambda_0 + 1.5)} = 0.05405\ldots$.

(iv) is a computation of the record: Remark 12.8.3 of `cycles.md`, by the script `experiments/uniform_trim.py`. The closing sentence is the content of (i)–(iii): at every period $p \geq 3$, and at every threshold for $p \geq p_1(X_0)$, the size conditions are met at the scale the trim allows, so no argument from size alone excludes cycles uniformly in p. □

The threshold, explicitly. $p_1(X_0)$ is explicit for each X_0 through the continued fraction of $\log_2 3$: $p_1 = 10$ for $X_0 \leq 16$, 22 for $2^8 \leq X_0 \leq 2^{12}$, 36 at 2^{20}, 98 at Simons and de Weger's $X_0 = 5 \cdot 2^{60}$ (consistent with their size-only exclusion of $m \leq 75$ there), 110 at 2^{71} and 309 at 2^{200}; over $X_0 = 2^j$, $j \leq 200$, $p_1 - \log_{\log_2 3} X_0 \in [2.784, 10.629]$ (`cycles.md` §12.8.7.4). The values at $X_0 = 2^j$, at $5 \cdot 2^{60}$ and at 2^{71} are computed by `experiments/staircase_x0_threshold.py`, and the two ranges follow from their endpoints because p_1 is nondecreasing in X_0 (Δ decreases as g grows). No closed form uniform in X_0 is claimed, and nothing is claimed below $p_1(X_0)$. At the threshold, Simons and de Weger's Lemma 7 reads $\Lambda < p\, c_p\, 2^{-\kappa_p n}$ [19, p. 58], against

the elementary $\Lambda < p/X_0$ of Corollary D.2; part (iii) is stated at the scale $\kappa_p n > g - \lambda_0 - \frac{1}{2}$, at which the two cross over (`cycles.md` Remark 12.8.7.6). There it attains the exponential dependence on the number of minima, and shows that the prefactor $p\,c_p$ cannot be lowered below 0.054. Whether it can be lowered *to* a constant — the $\log_2 p$ between parts (ii) and (iii) — is left open, as Simons and de Weger leave it [19, p. 69]. The integer profile of (iii) realizes the minimax value of Rozier's optimality remark under every rotation's exact size condition, $\kappa_p n \le \rho_0 \le \kappa_p n + \lambda_0 + 1.5$; what is attained is that value, not the inequality of his Lemme 4, whose additive term 2^p exceeds $2^{\kappa_p n}$ throughout this range (`cycles.md` Remark 12.8.7.6).

Remark D.7. The staircase profile — geometric depth growth with shallow exits — is exactly the growth regime of divergent orbits. Size analysis cannot forbid it as a cycle for the same reason drift analysis cannot forbid divergence. The two halves of the residual difficulty meet in one configuration.

Remark D.8 (the crash block). The crash block of the construction has depth 1 exactly when the budget binds, and depth at least 2 otherwise; the record's construction at the canonical scale (`cycles.md` §12.8.6.2) assumes a stronger hypothesis that forces depth 1. At the scale of Theorem D.6(iii) a unit block is excluded wherever the inequality below holds. If a size-passer with $p \ge 2$ has some $m_t = 1$, walking the recursion in the proof of Theorem D.5 from its reset, with $\mu_{j_1} = \mu_{j_1 - 1} + 1$ at the unit block j_1 in place of the block bound, gives $n < (\gamma + \log_2 p)\big((\log_2 3)^{p-1} - 1\big)/(\log_2 3 - 1) + (\log_2 3)^{p-1}$, that is,

$$\gamma + \log_2 p \;>\; \frac{(\log_2 3 - 1)\big(n - (\log_2 3)^{p-1}\big)}{(\log_2 3)^{p-1} - 1} \;\ge\; \log_2 3\,\kappa_p n - (\log_2 3 - 1)\,\frac{(\log_2 3)^{p-1}}{(\log_2 3)^{p-1} - 1}\,:$$

a unit block costs a factor $\log_2 3$ in the slope. Against $\gamma \le \kappa_p n + \lambda_0 + 1.5$ this excludes a unit block wherever $(\log_2 3 - 1)\kappa_p n \ge \lambda_0 + 1.5 + (\log_2 3 - 1)(\log_2 3)^{p-1}/((\log_2 3)^{p-1} - 1) + \log_2 p$, which at the periods $p_1(X_0)$ and $p_1(X_0) + 20$ holds from $X_0 = 2^{20}$ through 2^{200} (`cycles.md` Remark 12.8.7.5; the table is in the output of `experiments/staircase_x0_threshold.py`); below that the $\log_2 p$ slack leaves the unit crash undecided. Where the condition holds, the crash is forced to be deep: the last block continues the climb, as in Rozier's real extremal.

Verification and record. Prompted by correspondence with Eric Merle concerning the sharpness hedge of this theorem's statement in v1–v3 (his related formal work is [26]), parts (i)–(iii) were proved in the project record, `cycles.md` §12.8.6 and §12.8.7 at the commit of Appendix A, each checked there by independently written code with exact integer arithmetic at every size decision: part (i) by the scripts `experiments/staircase_gamma_upper.py` (45 checks, 0 failures), `experiments/staircase_allp_construction.py` and `experiments/staircase_allp_diophantine.py` (the sweep, the bracket, the window counts, and the construction at every period from 3 to 32 apart from $p = 4$, including the finite range $3 \le p \le 15$; the exhibitions at $p \in \{2, 4\}$ are by `experiments/staircase_allp.py`, §12.8.6.4), and parts (ii)–(iii) by `experiments/staircase_x0_threshold.py` (188 checks, 0 failures: the sandwich and the trim on 30,000 random configurations, the two-distance structure, $p_1(X_0)$ for $X_0 = 2^j$, $j \le 200$, and the construction end to end at 95 pairs (X_0, p)), and again by an independent re-implementation, `experiments/x0_threshold_independent_check.py` (43 checks, 0 failures). The proof printed above composes the record's availability theorem for (i) (§12.8.6.1) with its construction under the weaker hypothesis of §12.8.7.2. The configurations of (i) computed in the record all fail the divisibility conditions $q \mid R_r$, as those of (iv) do; those of (iii) were not

tested. Every statement here concerns the reach of size arguments, and none bears on whether cycles exist.

E The equidistribution hypothesis in full

This appendix states formally what Section 6 tells in words.

The window law

What is the right comparison object? By Theorem C.8 every dynamical decision at horizon one is a function of the depth-k window, so the null model is a measure on windows, and the structure forces the choice: the ω-residues carry the anchor digits, for which Haar measure is the canonical reference (this is where the classical 2-adic shift-conjugacy heuristics live), while the depth is *dynamical* — it is produced by Theorem C.5 — and receives instead its own exact renewal law, $d_+ = m_+ + a_+$ with m_+ geometric(1/2) and *independent* of the absorption a_+, so that it is the distribution of the depth, and not that of the absorption, which is the convolution of the two. The stationary 3-adic law governing a_+ (`aeh.md` §13.6.5) is not new here: it is Tao's Syracuse random variable $\mathrm{Syrac}(\mathbb{Z}_3)$ [4, Lemma 1.12 and Remark 1.13], in the present normalisation. Precisely, the 3-adic past-limit of the block coordinate has the law of $\mathrm{Syrac}(\mathbb{Z}_3)/2$ — the two differ by the 3-adic unit 2^{-1} because one block here is a run of Syracuse steps terminated by the first exponent ≥ 2, so that the block chain sees the Syracuse chain conditioned on that event — and hence $a_+ = v_3(\mathrm{Syrac}(\mathbb{Z}_3) + 2)$, with $P(a_+ = 0) = \frac{2}{3}$, $P(a_+ = 1) = \frac{19}{63}$ and $P(a_+ \geq 2) = \frac{2}{63}$ read off the nine values of $\mathrm{Syrac}(\mathbb{Z}/9\mathbb{Z})$ that Tao computes. What the present coordinates add is the *joint labelled* law (`aeh.md` §13.6.3(v)): the ω-residues are Haar-uniform among odd residues and independent of the depth, the whole carried on the residues *together with* the stratum labels (s, σ, a_+) — a_+ being a 3-adic function of ω (Lemma C.4) which the finite 2-adic residue window does not determine. Two integers fix the observable: a depth k and a *cap* D, chosen together and quantified over. The *capped depth-k window* at a visit is

$$W_{k,D} \;=\; \big(\,\omega \bmod 2^{k+2},\; d \wedge D,\; s \wedge D,\; \sigma \wedge D,\; a_+ \wedge D\,\big), \qquad u \wedge D := \min(u, D),$$

a finite alphabet of at most $2^{k+1}D^3(D+1)$ letters; the cap is what bounds it, and it must cap the labels as well as the depth, since σ and a_+ are unbounded. It is not the window of Theorem C.8, whose alphabet is countably infinite and which decides the next step: $W_{k,D}$ decides nothing, and no statement in this appendix asks it to. Let $\pi_{k,D}$ denote the law of $W_{k,D}$ under the two-sided Bernoulli measure $\hat{B} = \bigotimes_{i\in\mathbb{Z}}(\mathrm{geom}(\frac{1}{2}) \times \mathrm{geom}(\frac{1}{2}))$ on the door-letter alphabet — two-sided because the absorption a_+ is a function of the orbit's 3-adic past and has no realisation on the one-sided 2-adic side (`aeh.md` §13.6.3(iii), §13.6.5); reading the stationary Syracuse variable as the outcome of an iteration extending to arbitrarily large negative times is Tao's own alternative to his positive-time indexing [4, Remark 1.13, footnote 4 of arXiv v7]. "Product" names exactly two clauses of $\pi_{k,D}$ — residue $\perp$ depth, and $m_+ \perp a_+$ — and no others: $\pi_{k,D}$ is not a product across its coordinates (s is an exact function of the full state), and the process $(W_{k,D}(n))_n$ is stationary but not independent across time (e.g. from ($\omega \equiv 1$ (8), d odd) the next depth is exactly 1). Under $\pi_{k,D}$, unconditionally: $P(s = j) = 2^{-j}$ at every $j < D$, the mass beyond the cap sitting in one tail cell; the 3-gain rate is $\sum_{j\,\mathrm{even}} 2^{-j} = \frac{1}{3}$ by Lemma C.4, exact up to that same cell, whose parity split the cap hides; and the classical negative drift is a property of $\pi_{k,D}$ — but only of $\pi_{k,D}$: window equidistribution at each fixed (k, D) does not control the means

of the unbounded m_+ and s, so no drift or contraction statement about orbits follows from it (`aeh.md` §13.3.2). The folklore "ledger" is thus a theorem *about* $\pi_{k,D}$; the empirical question is only whether the orbits of typical starting values follow $\pi_{k,D}$.

The hypothesis

Hypothesis E.1 (AEH, ensemble form)**.** *Fix a* budget rate $\tau > 0$ *and a* block horizon rate $\theta > 0$*; for each* N *put* $b = \lfloor \log_2 N \rfloor$*,* $E_N = \lceil \tau b \rceil$ *and* $H_N = \lceil \theta b \rceil$*. Draw* x *uniformly from the odd integers of* $[N, 2N)$*; put* $(\omega_0, d_0) = R(x)$ *and* $(\omega_{n+1}, d_{n+1}) = F(\omega_n, d_n)$*; let the* letter *at block* n *be* $\ell_n = (m_{+,n},\, s_{n+1})$*, and let* $S_n = \sum_{i<n}(m_i + s_i)$ *be the* exponent spent *before block* n *— the number of* 2*'s divided out from* x *to* $x_{\mathrm{exit}}(n-1)$*. Tally block* n *at the symbol*

$$\tilde{\ell}_n \;=\; \ell_n \quad \textit{if } S_n < E_N, \qquad \tilde{\ell}_n \;=\; \dagger \quad \textit{otherwise.}$$

For a finite word $w = (w_1, \dots, w_L)$ *of letters, let* $f_N(w, x)$ *be the frequency of* w *among the* $H_N - L + 1$ *blocks* $(\tilde{\ell}_n, \dots, \tilde{\ell}_{n+L-1})$*,* $0 \le n \le H_N - L$*: every block counted exactly once, at weight* $1/(H_N - L + 1)$ *— the same number for every block of every orbit, so no block is reweighted by the orbit it came from and no denominator is random. Then for every finite word* w *and every* $\varepsilon > 0$*,*

$$\frac{2}{N}\,\#\big\{\, x \text{ odd},\ N \le x < 2N \ :\ \big|f_N(w,x) - B[w]\big| > \varepsilon \,\big\} \;\longrightarrow\; 0 \qquad (N \to \infty),$$

where, writing $w_i = (m_i, r_i)$ *for the components of the* i*-th letter of* w*,* $B[w] = \prod_i 2^{-(m_i + r_i)}$*, and* $B[w] = 0$ *for any* w *containing* †*, for every admissible pair* (τ, θ) *in the sense defined below.*

Equivalently, and this is the form the calibration measures: for every k, D and L, the empirical distribution of the L-blocks of consecutive capped windows $W_{k,D}$ over the same tallied blocks, with † carried through and given mass 0 by $\pi^{(L)}_{k,D}$, converges in total variation on the finite window alphabet, off a vanishing density of starts, to $\pi^{(L)}_{k,D}$, the L-block law of the stationary process under $\hat{B}$. The equivalence is Theorem 13.6.4 of `aeh.md`, a deterministic dictionary between letters and labelled window blocks; the case $L = 1$ recovers $\pi_{k,D}$, and $\pi^{(L)}_{k,D} \neq \pi^{\otimes L}_{k,D}$. A sampled segment has no infinite past at its first block, so the reconstruction of $W_{k,D}$ from letters is exact only away from the start; the deviation occupies $O(1)$ blocks of a horizon $H_N \to \infty$ and contributes $O(1/H_N)$ to every frequency above.

There is one limit here, $N \to \infty$, and the sample grows because the sampling scale grows rather than because any one orbit is run forever. That is forced: along a fixed convergent orbit the unrestricted empirical distribution is false, the tail sitting at $(1, 1)$ forever, while above a fixed cut such an orbit supplies only finitely many qualifying visits and none at all once the cut exceeds its maximum, so a limit in orbit length is empty rather than merely delicate, in whichever order it is taken (`aeh.md` §13.6.6).

The budget and admissibility

The horizon does the job a bulk cut would do, and does it without one. A step of the one-division map $y \mapsto y/2$ or $(3y+1)/2$ never lowers $\log_2$ by more than 1, so $\log_2 x_{\mathrm{exit}}(n-1) \ge \log_2 x - S_n$ for every start and every n, with no exceptional set; inside a budget $\tau < 1$ of the start's own bits every tallied exit therefore exceeds $N^{1-\tau}$, far above the *bottom regime* of `aeh.md` §13.1, the fixed

drainage basin of small integers. That matters because any altitude threshold is a selection on the observable itself — a cut on $x_{\text{exit}} = (3^d\omega - 1)/2^s$ censors large s, and a cut on the core ω_+ censors large s, m_+ and a_+ together — whereas the budget clause $S_n < E_N$ is *predictable*, decided by blocks strictly earlier than the one it admits, and the tally denominator is the deterministic H_N that `aeh.md` §13.5's standing rule was written to secure. Call τ *protected* when all but a vanishing density of starts keep every in-budget exit above any X_N with $\log X_N = o(\log N)$, and (τ, θ) *consistent* when $\theta\,\mathbb{E}_B[m+r] < \tau$; *admissible* means both. Every $\tau < 1$ is protected outright by the bound above, and every $\tau < (1 - \log_2\sqrt{3})^{-1} = 4.8188\ldots$ is protected at natural density one by Inselmann [5, Thm. 1.1], whose two-sided envelope is stated for the one-division map — the same unit, nothing converted. Consistency is compatibility with the target law rather than a claim about orbits: under B a block spends $\mathbb{E}_B[m+r] = 4$ of exponent, so $4\theta < \tau$ says exactly that B itself predicts no out-of-budget block, which is what $\pi_{k,D}$'s giving † no mass requires. Where the cylinder count runs it is more, and unconditionally: for $\tau < 1$ no block of the horizon leaves the budget and the empirical exponent mean converges to 4 (`aeh.md` Lemma 13.2.4(g), whose two error terms are, on that range, precisely the two clauses of admissibility). Past that range the hypothesis supplies less, and only about frequencies: a vanishing frequency of † says all but $o(H_N)$ of the first H_N blocks are within budget, and is not a bound on a sum over its complement — the block at which the budget is exhausted is tallied at full weight and its letter is unbounded, and the $o(H_N)$ blocks past it are uncontrolled, so $H_N^{-1}\sum_{n<H_N}(m_n + r_n) \to 4$ does not follow there and can fail by any amount: a single letter of size cH_N at the exhausting block, with the $o(H_N)$ blocks after it tallied at †, leaves every fixed pattern frequency at its B-value and moves the mean from 4 to $4 + c$ (`aeh.md` §13.2). Hence $4\theta < \tau < 1$ is $\theta < 1/4$ and $4\theta < \tau < 4.8188\ldots$ is $\theta < 1/\beta = 1.2047\ldots$, where $\beta = 2(2 - \log_2 3) = 0.8301\ldots$ is the classical per-block contraction rate and $4/\beta = (1 - \log_2\sqrt{3})^{-1}$ an identity — arithmetic on the definition of consistency, with the divisor in either block reading the mean of the target law: a theorem about orbits below the budget, and not a consequence of the hypothesis above it.

The unconditional base case

The hypothesis has an unconditional base case, and it is Heuristic C.9 with a number on it. The classical coding fact — the odd integers whose first n blocks realize a prescribed itinerary form exactly one residue class modulo 2^{S+1}, S the itinerary's total exponent (Terras [1]; in the present coordinates `itinerary.md` §14.15.1.5) — makes the first n blocks of a start drawn uniformly from *any* window of 2^J consecutive integers exactly product-distributed on the event $S + 1 \le J$: the event that the itinerary has not outspent the start's supply of binary digits. That fact is about the word beginning at x itself, while ℓ_0 is the letter of the block *after* the start's own; it is applied here to the *extended* $(n+1)$-letter word — the start's own letter followed by $\ell_0, \ldots, \ell_{n-1}$, of total exponent exactly S_{n+1}, the divisions from x down to $x_{\text{exit}}(n)$ — of whose law the one displayed is a marginal. Writing $b = \lfloor \log_2 N \rfloor$, this gives, for every $J \ge 2$,

$$\big\|\mathrm{Law}(\ell_0, \ldots, \ell_{n-1}) - B^{\otimes n}\big\|_{\mathrm{TV}} \;\le\; \frac{2^{J+2}}{N} + P_B(S_{n+1} \ge J),$$
$$P_B(S_{n+1} \ge J) = P\big(\mathrm{Bin}(J-1, \tfrac{1}{2}) < 2(n+1)\big),$$

the second identity because S_{n+1} is the waiting time for the $2(n+1)$-th head in a fair coin sequence. The first term is the price of a general window $[N, 2N)$ over a dyadic one and is negligible for $J \le (1-\zeta)b$, $\zeta > 0$ fixed; the second decays at the exact rate $\log 2 - \mathcal{H}(2\theta)$ per bit

at $n = \lceil \theta b \rceil$, $\mathcal{H}$ the binary entropy — positive for every $\theta < 1/4$ and *zero* at $\theta = 1/4$ (`aeh.md` §13.2). The bound is on the joint law of the whole length-n word, so it controls every finite block length at once, and with the concentration of empirical pattern frequencies, the removal of the segment's initial past-boundary and the altitude bound above — which makes the budget its own protection — it yields the hypothesis outright: Hypothesis E.1 is a *theorem* for every admissible (τ, θ) with $\tau < 1$, equivalently for every horizon rate $\theta < 1/4$, at every block length. The assembly is Lemma 13.2.4 of `aeh.md`.

The natural clock here is total exponent, not blocks: the frontier is one unit of exponent per bit of start — the start's own digit supply, with no estimate in it — and it reads as $1/4$ blocks per bit only through the mean exponent per block, $\mathbb{E}_B[m + r] = 4$, a conversion available here precisely because the word is exactly B here. Hypothesis E.1 is the assertion that the law still describes the orbit after the start's digits have been spent; the digit budget locates the frontier, and this is the statistical statement on the far side of it. That frontier is the classical one — the same $\theta < 1/4$ is where Terras's count stops, and $1 - \beta/4 = \log_4 3$ is exactly the exponent Korec [3] obtains by exhausting it — and it is a frontier of this technique and not of the problem, Inselmann [5] crossing it unconditionally for the trajectory envelope and the first moment, in step time and by a stronger density notion rather than a sharper count (`aeh.md` §13.3.2).

What the hypothesis implies, and what it does not

AEH implies the ledger, in the form the hypothesis has: s is a coordinate of the observable, so $P(s = j) = 2^{-j}$ is a marginal of $\pi_{k,D}$ rather than a deduction, exactly at every $j < D$ and up to the cap's single tail cell above; and for every ε and every horizon rate, all but a vanishing density of the odd starting values of each size carry those frequencies along their first H_N blocks, all but $o(H_N)$ of them within the budget — and all of them when $\tau < 1$. Evaluating each odd x once, at its own dyadic scale $2^{\lfloor \log_2 x \rfloor}$, turns that array of statements into a single one: the odd integers that fail at their own scale have natural density zero (`aeh.md` Proposition 13.2.5). The union over *all* sampling scales is a different and larger object and is not controlled — the bad sets form a triangular array, each testing a horizon and a budget that move with the scale, and vanishing density in every window does not thin their unrestricted union. The descent consequence is not stated here, because it is a theorem without the hypothesis and a stronger one: Inselmann's [5, Cor. 1.4, Thm. 1.10], cited in Appendix F, are unconditional and run to $4.8188\ldots$ times the classical range at which this appendix's cylinder count stops, the two measured in the same units, the second two-sided and uniform in the time. His [5, Thm. 1.6], a density statement about odd steps, is what carries his own passage from one-division-map time to Syracuse time; the further passage to block time needs the frequency with which a Syracuse step ends a block, a two-letter statistic of the parity word that neither theorem supplies and that $\pi_{k,D}$ does, so it is not available to underwrite the hypothesis: inside the digit budget it is a theorem of the cylinder count (`aeh.md` Lemma 13.2.4(g)), and past it neither a theorem nor a consequence of the hypothesis (`aeh.md` §13.2.3). What the hypothesis supplies beyond these is distributional and is the whole of its content: the full 2^{-j} marginal at every j below the cap, the exact $\frac{1}{3}$ rate of Lemma C.4, and the depth law, at every finite depth and past the budget at which the classical count stops. Its exceptional set is a set of *starting values* of vanishing density at a prescribed finite horizon — not a null set of orbits, and not a statement about any orbit's infinite tail. It does not exclude individual staircase tails (Remark D.7); it does not iterate — the image of a density-one set of starts need not be density-one at the next scale, an obstruction Tao states in the same terms [4]; and, by Heuristic C.9, it is not expected to be reachable by finite-window

computation. Its content is a question about digits of 2-adic logarithms, older and broader than the Collatz problem.

Calibration

An eight-round campaign (per-orbit statistics over each cell's own conditional counts, 1,600–2,600 independent orbits per cell, under an altitude cut on the core) produced four apparent violations of $\pi_{k,D}$-uniformity, each dissolved by a sharper control: (1) an s=3 marginal excess and (2) pair correlations, both artifacts of small starting values; (3) digit biases, contamination by the bottom regime, where deviations reach $z = 41$ but reflect particular numbers rather than a measure; and (4) a 5σ digit bias in one cell surviving a 2^{40} cut (code: `experiments/aeh_calibration.py`, and for the fourth `experiments/aeh_anomaly.py`). The last dissolved under fixed-horizon unweighted sampling (0.2503 ± 0.0015 against null 0.25 over 84,739 visits) and was explained exactly:

Lemma E.2 (deterministic routing)**.** *For $\omega \equiv 1 \pmod 8$, $d = 1$: here $s = 1$, $C = 3\omega + 1$, $v_2(C) = 2$, $a_+ = 0$, so $\omega_+ = (3\omega + 1)/4$ and $d_+ = 1$; the residue $\omega \bmod 32$ determines the next class exactly: $1 \mapsto [1, 1]$ (immediate return to the same class), $9 \mapsto [7, 1]$, $17 \mapsto [5, 1]$, $25 \mapsto [3, 1]$, the lifting class. Here $[r, 1]$ denotes the class $\omega \equiv r \pmod 8$, $d = 1$, not a state.*

Ratio estimators over correlated visit sequences inherit this routing (residue 25 leads into deep descent, hence fewer qualifying visits, hence inflated per-orbit weight), manufacturing a bias aimed at one residue of the class $[1, 1]$, whose residue 1 returns to it deterministically (a self-loop channel). No residual discrepancy was detected at any tested depth or cell, under the stated protocol and within three limits the campaign does not reach past: it tests block lengths $L = 1$ and $L = 2$ only and is silent above; its adjudicating runs pool per visit across orbits, so they measure the across-orbit average of the per-start frequencies — a consequence of Hypothesis E.1 rather than the per-start statement itself, which would be tested by the distribution across orbits of $\|\nu - \pi_{k,D}\|_{\mathrm{TV}}$ and is not among the runs reported; and its altitude guard cuts on the core ω_+ rather than the door, censoring s, m_+ and a_+ at once, and does bind at finite size, removing 2.6% of visits in the adjudicating run and moving the reported statistics at the third decimal (`aeh.md` §13.4, §13.5). Hypothesis E.1 carries no cut of any kind; it tallies within the exponent budget, which bounds every tallied exit from below without selecting on the letter being tallied.

F Related work in detail

Block-like decompositions and stochastic models are classical (Terras [1]; see Lagarias [21] for the literature), and the 2-adic conjugacy of the Collatz map to the shift underlies all Haar-genericity heuristics. The unconditional density line descending from Terras is long and is not what Appendix E adds to. Terras [1] and Everett [2] place descent below the start at natural density one; Korec [3] lowers the target to y^c for every $c > \log_4 3 = 0.7924\ldots$, in natural density and within $O(\log y)$ steps; Inselmann [5] carries a two-sided trajectory law — the orbit tracked to within $m^{\pm\varepsilon}$ of its classical drift, simultaneously at every time up to $(\log_2 \frac{4}{3})^{-1} \log_2 m$ Syracuse steps, the full descent horizon in that unit — at natural density one, and so reaches m^{ε} for every fixed $\varepsilon > 0$; and Tao [4] reaches every divergent threshold, in logarithmic density. All of these horizons are counted in steps, and their reading in the reduced blocks of Appendix E is not free: it divides by a mean number of steps per block which is a theorem of the cylinder count inside

the digit budget and, past it, neither a theorem nor a consequence of Hypothesis E.1. What Hypothesis E.1 asserts is not descent but the letter statistics themselves — equivalently the exact window law $\pi_{k,D}$ — at horizons past the digit budget of Heuristic C.9. Tao's paper carries a footnote to two further 3-adic studies, and neither is the object appearing here. Wirsching [6] completes the root of the predecessor tree 3-adically and studies the inverse map as a Markov chain on $[0,1] \times \mathbb{Z}_3^\times$ in order to *count* predecessors; his inverse branches therefore carry equal weight and his limiting 3-adic law is Haar, whereas the law appearing in Appendix E weights each branch by the probability 2^{-a} that an orbit takes it and is consequently not uniform — $(\frac{2}{3}, \frac{1}{3})$ at level one where Haar gives $(\frac{1}{2}, \frac{1}{2})$. The two programmes are nonetheless siblings in shape: what Wirsching needs is that a Collatz-generated object be unbiased in 3-adic coordinates, and Hypothesis E.1 is that one is unbiased in 2-adic coordinates. Thomas [7] proves that the integers whose orbit reaches 1 with all six frequencies over the units modulo 18 below $\frac{1}{6} + 0.0215$ have counting function at most $N^{0.9999}$; he constructs no limiting measure. The stationary 3-adic law of Appendix E (Tao's Syrac) is far from the near-uniformity he rules out: at level one it puts mass $\frac{2}{3}$ on the residue 2 mod 3, which three of his six classes share, so a sequence with those frequencies has some class at frequency at least $\frac{2}{9} > \frac{1}{6} + 0.0215$. That his counting bound follows from the law is not claimed. Two-adic and Diophantine methods dominate the cycle literature [8, 9, 19, 20, 18]; effective linear forms in logarithms [22, 23, 24] are the engine, and the verified range of the conjecture stands at 2^{71} [25]. The linear cycle equation behind Proposition D.1 is Böhm and Sontacchi's [11].

The trim of Theorem D.5 is classical. For every number of local minima of a cycle — which is our number of blocks p (Appendix D) — Rozier [10, Lemme 4] bounded the number of odd steps, with the exponent $(\log_2 3 - 1)/((\log_2 3)^p - 1)$; Brox [13] proved finiteness for each p; and Simons and de Weger [19, Lemmas 7, 14 and Theorem 3] made the bound explicit and computed with it. Our proof is a rederivation by another route. Rozier observed that his bound is optimal over real run-length tuples, the extremal tuple growing geometrically at ratio $\log_2 3$ [10, p. 11]; Simons and de Weger expected the exponential dependence on the number of minima to be unavoidable [19, p. 69]. In the cycle-length parameter, Halbeisen and Hungerbühler [12] proved the size criterion optimal, attained by explicit rational cycles of every length, and reduced the remaining question to divisibility [12, p. 238]. Theorem D.6 gives integer configurations meeting every rotation's exact size condition at every period $p \geq 3$, and, for $p \geq p_1(X_0)$, with every element at least a threshold X_0; at that threshold scale they realize the minimax value of Rozier's extremal — the sharpness Simons and de Weger expected.

G Verification index

Every computational claim of the paper, with the check that supports it as stated where the claim is made. "Record" is a section of the project record at the commit of Appendix A; scripts are in its `experiments/` directory and run as `python experiments/<name>.py`.

Statement	What was checked	Script; record
Theorem C.3, Lemma C.4, Theorem C.5 (Remark C.6)	Independently written code, zero discrepancies: the valuation law on 8,000 random states across both lifting components; the entry-depth law exhaustively on all 62,937 valid states with $\omega < 3000$, $d < 64$; the absorption law on 327,980 states (983,954 comparisons), with the branches $d = h(s)$ and $d < h(s)$ constructed rather than sampled	`absorption_law.py`; `stage3.md` §11.8.6.3 and its status header
Theorem C.7 (Remark C.10)	The low-order anchor-increment law	`anchor_increment.py`
Theorem C.8 (Remark C.10)	Along real orbits, $k \in \{4, 8\}$: 21,296 steps, 21,000 decided with zero errors, 296 flagged deep with zero violations of $s_+ \geq k + 2$	`one_step_propagation.py`
Undecided rate (after Theorem C.8)	Measured 0.0275 against 0.03125 at $k = 4$, 0.0019 against 0.00195 at $k = 8$	`stage4.md`, Remark "undecided rate"
Lemma D.3, $p \leq 5$	The odd values $1, 3, 5, 7$ reach 1; the orbits are printed in the proof	by inspection
Theorem D.4	Exact searches: $p = 1$ for $m \leq 20{,}000$; $p = 2, 3$ to $n = 20{,}000$; no solutions	`period1_cycles.py`, `period2_cycles.py`, `period3_cycles.py`; `cycles.md` §12.2.3, §12.5.3, §12.7.5
Theorem D.6(i)	45 checks, 0 failures; the sweep, the bracket, the window counts, and the construction at every period from 3 to 32 apart from $p = 4$, including $3 \leq p \leq 15$; exact integer arithmetic at every size decision	`staircase_gamma_upper.py`; `staircase_allp_construction.py`, `staircase_allp_diophantine.py`; `cycles.md` §12.8.6
Theorem D.6(i), $p \in \{2, 4\}$	The exhibitions	`staircase_allp.py`; `cycles.md` §12.8.6.4
Theorem D.6(ii)–(iii)	188 checks, 0 failures: the sandwich and the trim on 30,000 random configurations, the two-distance structure, $p_1(X_0)$ for $X_0 = 2^j$, $j \leq 200$, the construction end to end at 95 pairs (X_0, p); and an independent re-implementation, 43 checks, 0 failures	`staircase_x0_threshold.py`; `x0_threshold_independent_check.py`; `cycles.md` §12.8.7
The values of $p_1(X_0)$ ("The threshold, explicitly")	The values at $X_0 = 2^j$, at $5 \cdot 2^{60}$ and at 2^{71}; the ranges follow by monotonicity	`staircase_x0_threshold.py`; `cycles.md` §12.8.7.4
Remark D.8	The table of the unit-block condition at $p_1(X_0)$ and $p_1(X_0) + 20$, $2^{20} \leq X_0 \leq 2^{200}$	output of `staircase_x0_threshold.py`; `cycles.md` Remark 12.8.7.5
Theorem D.6(iv)	The $p = 7$ staircase and the 84 instances at $p = 6$	`uniform_trim.py`; `cycles.md` Remark 12.8.3
Calibration (Appendix E)	The eight-round campaign; the fourth anomaly	`aeh_calibration.py`; `aeh_anomaly.py`; `aeh.md` §13.4, §13.5

Statements proved in the record and not printed here. The paper names the record section for each: the second display of Theorem D.5 (`cycles.md` Corollary 12.8.2, after [19, Lemma 14, Theorem 3(d)]); the details of Theorem D.4 (the `cycles.md` sections named in its outline); the comparisons of "The threshold, explicitly" (`cycles.md` Remark 12.8.7.6); the joint labelled law and the two-sided measure (`aeh.md` §13.6.3, §13.6.5); the equivalent form of Hypothesis E.1 (`aeh.md` Theorem 13.6.4); the single-limit argument (`aeh.md` §13.6.6); the base case and the exponent mean (`aeh.md` Lemma 13.2.4, with §13.2.3); and the dyadic-scale density statement (`aeh.md` Proposition 13.2.5).

H Concordance with v3, and notation changes

v3 numbers its statements by section; v4 moves every formal statement into an appendix, in the same order, so that a v3 number becomes a v4 number by a change of the leading section number to a letter.

v3	v4		v3	v4	
Def 2.1	Def B.1	reduced map	Prop 4.1	Prop D.1	elimination
Prop 2.2	Prop B.2	blocks	Cor 4.2	Cor D.2	size balance
Thm 2.3	Thm B.3	equivalence	Lemma 4.3	Lemma D.3	ceiling
Def 3.1	Def C.1	anchor	Thm 4.4	Thm D.4	periods 1–3
Lemma 3.2	Lemma C.2	squaring	Thm 4.5	Thm D.5	uniform trim
Thm 3.3	Thm C.3	valuation law	Thm 4.6	Thm D.6	staircase
Lemma 3.4	Lemma C.4	absorption	Remark 4.7	Remark D.7	divergence
Thm 3.5	Thm C.5	depth closure	—	Remark D.8	crash (new)
Remark 3.6	Remark C.6	checks	Hyp 5.1	Hyp E.1	AEH
Thm 3.7	Thm C.7	increment	Lemma 5.2	Lemma E.2	routing
Thm 3.8	Thm C.8	trichotomy	eq. (1)	eq. (1)	isometry
Heur. 3.9	Heur. C.9	digit budget	—	eqs. (2), (3)	new
Remark 3.10	Remark C.10	checks			

v3's sections map as follows: §1 (introduction, related work and provenance) to Sections 1 and 8 and Appendix F; §2 to Sections 2–3 and Appendix B; §3 to Section 4 and Appendix C; §4 to Section 5 and Appendix D; §5 to Section 6 and Appendix E; §6 (discussion) to Section 7; Appendix A to Appendix A.

Renamed symbols. Symbols that carried two meanings in v3, or would have in v4, are renamed here; no meaning changed. (η now names only the quantity $\phi(s_c)$ in the proof of Theorem D.6.)

v3	v4	meaning
ε (Theorem 3.5)	ξ (Theorem C.5)	the anchor displacement
ε_t (Corollary 4.2, Lemma 4.3)	χ_t (Corollary D.2, Lemma D.3)	the block ratio $(2^{s_t}-1)/(3^{d_t}\omega_t)$
T_j (proof of Theorem 4.5)	μ_j (also in the proof of Theorem D.6 and Remark D.8)	partial sums of entry depths
—	s_{tot} (proof of Theorem D.6, new in v4)	$K-n$, the total exit valuation
T_N (Hypothesis 5.1)	H_N	the block horizon
Λ_N (Hypothesis 5.1)	E_N	the exponent budget
H (§5)	$\mathcal{H}$	the binary entropy
η (§5)	ζ	a fixed positive slack
$(r,1)$ (Lemma 5.2)	$[r,1]$	the class $\omega \equiv r \pmod 8$, $d=1$

Unchanged: T is the odd-to-odd map throughout; Λ is the linear form $K\log 2 - n\log 3$; ε is a tolerance; M_t and S_t (Proposition D.1) keep the record's notation.

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